Rank Fine, Pack Fine, Call Nothing

79.09% on the benchmark Mem0 published, at a sixth of the tokens, with no generative calls

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Executive summary

What this is. A memory layer for long conversations. Every turn, it decides what the model should be reminded of, and fills a fixed amount of space with it. It does that **without ever calling a language model to help** — no summarizing, no note-taking, no rewriting.

The design copies three things human memory does, and runs all three at once:

- **Recency — what just happened.** The most recent exchanges always go in, automatically. You don't search your memory for what someone said a minute ago; it is simply still there.
- **Depth — what this reminds it of.** Every exchange ever had is kept word for word and indexed by meaning, so mentioning *my sister's wedding* can pull back a conversation from six months ago. This is recall by cue, and it is the part most memory systems stop at.
- **Spread — covering ground instead of repeating it.** This is the part that makes the design different, and it is easiest to see by what goes wrong without it. Rank every past exchange by how well it matches the question, take the best ten, and you often get the same fact ten times — ten occasions the person mentioned their sister. The space is full and the model learned one thing. Instead, each slot is filled by asking a different question: *given what has already been picked, what does this one add?* Ten slots end up holding ten different things.

Everything is delivered exactly as it was said. Nothing is summarized or rewritten, so nothing the model is told about the past can be wrong.

Where it lands. The evaluation harness behind arXiv:2504.19413's Table 2 — question set, answer prompt, judge prompt, judge model and metric — was rebuilt here, and this component run through it on all 1,540 scored LoCoMo questions.

System	LLM-as-a-Judge	Mean prompt tokens
This component	**79.09%**	**4,243**
Full context	72.90%	—
Full context, reproduced here	*72.47%*	*25,405*
Mem0^g	68.44%	—
Mem0	66.88%	—
Zep	65.99%	—
RAG, best variant	60.53%	—
LangMem	58.10%	—
OpenAI memory	52.90%	—
A-MEM	48.38%	—
No memory at all	**26.30%**	84

79.09%, on a sixth of the tokens the table's previous best spends. The highest row there is full context at 72.90%, and this rig reproduced that row at **72.47% — a gap of 0.43 points**, so the comparison rests on a control it verified rather than an assumption. Eight rows are quoted from Table 2 and were not re-run here; nothing in this paper is tested against them.

The result the study **registered in advance** cleared its bar by more. Against fixed-chunk retrieval: **+33.31 points, 558 items won against 45, p = 6.615e-114**, with a second endpoint that uses no model at all agreeing.

Abstract

We report a deterministic memory layer for long conversations that makes no generative model calls, and the pre-registered programme that reduced it to four components: ten numbered studies plus one registered exploratory bakeoff. The surviving design is an append-only verbatim store, a recency window, cosine-threshold similarity retrieval, and a set-level coverage objective, packed against one character budget at exact serialized cost.

We rebuilt the evaluation harness behind arXiv:2504.19413's Table 2 — its question set, answer prompt, judge prompt, judge model and metric — and ran this layer through it on all 1,540 scored LoCoMo questions. It scores 79.09%, above every row of that table, on 4,243 prompt tokens against the 25,405 of full context reproduced here. None of the systems behind those rows was re-run. The placement is licensed by reproducing their own ceiling row first: full context, the configuration with nothing to choose, came back at 72.47% against a published 72.90%, and the judge moved 0.06 points across two scorings of the same sealed answers. Six of the table's rows were not re-run and are quoted with attribution; no test here is computed against them.

Two registered gates failed and both are reported as results. With no memory block at all the reader still scores 26.30% against a bar of 5%, because the judge prompt instructs the grader to accept any answer touching the gold answer's topic — 32.34% on open-domain, the largest of four strata. That floor is a property of the shared instrument and the source paper does not report it. The second gate missed by 14.75 points: the sweep behind Table 2's "RAG (best variant)" row spans 26 points with the published value inside it, so one configuration choice moves that row further than the distance separating most rows on the table.

Against Mem0 2.0.18, installed and run here on one local reader at a matched 16,000-character budget, the layer that makes no generative calls answers 7.7 points more of 300 questions than the layer that spent 1,646 of them across 284 minutes (46 gains against 23, one-sided exact p = 0.0038; a model-free containment endpoint agrees at +9.7 points, p = 2.85e-05). It assembles a context block in 10 milliseconds against 413, in a store of 7.2 MB against 42.8. Of the answers stated verbatim in those conversations, up to 21% are absent from Mem0's store, an upper bound because a preserved paraphrase counts as absent; 31% of message pairs produced no memory at all. Per query Mem0 is the cheaper arm, at 3,392 prompt tokens against 4,009, so ingest cost and read cost run in opposite directions and both are reported. LoCoMo fits the reader's window, so both studies measure cost and accuracy rather than capacity.

The programme's central positive result is a granularity rule confirmed on a sealed external holdout. On six LoCoMo conversations withheld until bars, endpoint and budget were locked, ranking adjacent-turn pairs by their own cosine raises complete exact-evidence delivery from 843 to 935 of 1,098 over assigning each pair its session's maximum score, with 140 gains against 48 losses (ratio 2.92, one-sided exact p = 6.19e-12) and every conversation net positive; source order reaches 258. The result is bounded to evidence availability and authorizes no reader claim.

Two further corpora locate the mechanism. On 465 turn-labelled LongMemEval questions, own-turn ranking raises any-exact-evidence delivery from 361 to 461 with 100 gains and no losses; evidence turns have median 298 characters against 2,550 for their parent episodes, and parent length correlates with worse normalized own-cosine rank at Spearman rho 0.484. On the internal 121-turn store the same substitution takes an enumeration probe from 12 to 14 of 17 with 21 of 21 targeted items preserved. The relation is not monotone in granularity: on the same corpus, ranking at the episode rather than the session loses 37 items, so the episode is worse than both its parent and its child. The rule is therefore stated conditionally — rank at a unit small enough that its embedding is dominated by the material answering the query, and pack at the finest affordable unit — and that first clause has no operational test here. Splitting also changes character count and semantic localization together, so length is not isolated.

Underneath sits a decomposition of retrieval failure into three constraints that bind in a forced order. The candidate pool binds first and structurally — one of four domains has no representative anywhere in the deployed shortlist, so no selection rule reaches it from there. The objective binds second and only after: run on the deployed pool, the shipped objective scores below the baseline it replaces. A similarity floor binds last, at a query cosine no reweighting closes. Capacity was never the constraint: an exact optimum computed with the answer key makes 14 of 17 items available in 5,058 of 32,000 characters while deployed selection made 6 available and spent 31,946.

We report the negative results at the same resolution. Write-time selection failed in five studies because it cannot anticipate a later query; moving selection to query time failed in a registered bakeoff; graph construction, query routing, approximate search, query segmentation and attention-derived term selection each failed their own gates. Three sealed-holdout experiments returned no signal at all. We also report the instrument: five replicates under identical conditions scored 11.0, 8.0, 8.0, 8.0 and 8.0 in that order, so the

run-to-run band is 3.0 points and three of the arc's four scored verdicts fall inside it and are labelled not demonstrated. The divergent run was the first in a fresh server process. The offline delivery results are counts and identities and are unaffected.

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1. Introduction

A long conversation forces a trade. Keep the whole transcript and the model slows down and loses the middle. Summarize it and the details are gone. A third option is to store every exchange verbatim and rebuild a small, relevant context each turn, which moves the problem from compression to selection.

This paper is about that third option, built and measured over eleven pre-registered efforts — **ten numbered studies plus one registered exploratory bakeoff**, counted that way throughout. Most of the mechanisms failed. What survived is smaller than what the programme set out to build, and the reason it is worth reading is that the survivor is now confirmed on a corpus this programme did not construct, and that the failures are specific enough to name.

The one-sentence result: **the unit you score a candidate by is a bigger lever than the rule you score it with, and neither requires a language model** — with §7.3 the counterexample that stops "smaller is better" from being the reading.

1.1 What kind of paper this is

A single-programme experience report with one external confirmation. One internal corpus, one rubric locked since the second study, one local model at one quantization, one machine, one seed. Every *scored* comparison is a single run, and §13.1 gives the measured variance that bounds them.

That scope applies to the scored results. It does not apply to the deterministic ones, and the distinction decides most of what this paper is permitted to say. The delivery counts in §6 and §7 are produced with zero model calls, reproduce byte-identically on replay, and are verified against committed hashes. They are counts and identities rather than judgements. Where a result is one of those, this paper states it plainly; where it is a score, it carries its band.

1.2 Contributions

1. **A granularity rule with a sealed external confirmation** (§6, §7). Ranking at the finest informative unit, confirmed prospectively on six withheld LoCoMo conversations and reproduced on two further corpora with the size mechanism measured rather than asserted.
2. **A decomposition of where retrieval failure lives** (§8): a candidate pool, a selection objective, and a similarity floor. They are not independent — they bind in a forced order, and applying the second fix without the first makes the shipped configuration worse than the baseline it replaces.
3. **The measurement that makes the decomposition possible**: a per-fact known optimum computed on the same store under exact serialized-cost accounting (§8.1). It costs an answer key and exact cost accounting, which is why it is unusual rather than difficult.
4. **A subtraction result** (§11): twelve mechanisms removed, each by its own gate, and the argument that the properties making the survivor deployable followed from the removals.
5. **A correction record** (§12) including the instrument's own measured noise band, and one case where a diagnostic written to catch a specific failure class nearly committed that exact failure.

1.3 What this paper does not claim

No scored difference below about three points in this arc is claimed as real; §13.1 gives the measurement. Mem0 was run here and §5 reports it; no comparison against HippoRAG, Zep or Letta was run, and §2 says exactly what is and is not being compared. No general claim that similarity retrieval fails — §8.4 measures the opposite on eight of nine internal probes and §9 measures it on 470 external ones. No novelty for maximal marginal relevance, facility location or submodular selection, which are established methods; what is offered is the decomposition and the measurement, not the selector. And no claim that the internal 12 of 17 is good: it sits below the programme's registered bar of 14, and below the 15 a known optimum reaches on the same store for a sixth of the cost.

2. Related work and positioning

Studies 001–010 were designed and run before this programme's first literature scan, and are committed with SHAs that show it. We state that once, because it explains why early designs rediscovered known ideas, and press it no further: arriving independently at a worse version of a published method is not a contribution.

Entity-centric indexing. HippoRAG builds a knowledge graph over extracted entities and retrieves by traversal; the authors' own follow-up identifies entity-centricity as a limitation. This programme's hardest repeated failure is consistent with that. Six target facts sit in spans where the entity extractor finds zero entities, so an entity-gated index has no path to them, which is why entity extraction was ruled out as a primary index here.

Structure over retrieved units. GraphRAG, SGMem and CodaRAG impose explicit structure on retrieved material. This programme built the corresponding mechanism — an associative graph over observed co-activation — and no configuration cleared its advancement gate (§11).

Deployed memory systems. Letta, Mem0 and Zep ship in this space. Full citation detail, published numbers and verification status are in paper/notes/COMPETITIVE_LANDSCAPE.md.

2.1 What is and is not being compared

Of the systems named above, only Mem0 was run here — §5. Every number attributed to any of the others is cited from its publication and labelled as such.

The measures differ, and for most of this programme's life that difference was the obstacle. Mem0 and its neighbours report LLM-judged question-answering accuracy on LoCoMo. §6 and §7 report deterministic evidence *availability* at a fixed character budget: whether the text carrying an answer was present in the delivered context, established without a model in the loop. **Those two are never placed in one column.** Doing so would be exactly the substitution this programme's operating manual names as its recurring failure — *a surrogate that can pass without the property it claims to certify.*

§5 does place a number beside theirs, and it is admissible because it is not a surrogate: HH-002 measures LLM-judged accuracy itself, on their harness, their questions and their judge. The substitution above stays forbidden; what changed is that this programme no longer needs it. NF-004's 935 of 1,098 is an availability count and never enters that table.

What is comparable is architectural, and the axes are countable from either system's own description:

Axis	This component	Systems that spend a call on this layer
Generative calls per stored turn	**0**	Mem0's ingestion is one extraction call per message pair **plus one update call per extracted fact**, so 1 + n Graphiti runs entity extraction, entity resolution, fact extraction, temporal extraction and edge invalidation per episode
Delivered text	Stored episodes verbatim	Model-written summaries or extracted facts
Replayability	context() byte-identical across processes; 132 payloads reproduce by SHA-256	Bounded by generation determinism
Failure mode	An episode is not delivered	An episode is not delivered, or is delivered as a wrong paraphrase
What is measured	Evidence availability	Judged answer accuracy

Only the first row is arithmetic on published descriptions. It is the one comparison this paper can make without running anything, and the axis on which the difference is largest. The remaining rows are not of that kind and should not inherit its licence: rows two and three describe *this* component and are measured here, with the competitor column stating what follows from a generative pipeline in principle rather than anything observed. Row four's "delivered as a wrong paraphrase" is a failure mode those architectures admit, not one seen in a run. Row five is a definition.

One collision is worth naming explicitly, because it is where a reader is most likely to construct a sentence this paper refuses. Both this programme and Mem0 report a number "on LoCoMo". §6.1 ran six sealed conversations with zero model calls during measurement and counts whether evidence text was delivered. Mem0's LoCoMo figure has a model answering and a model judging. The corpus name is shared; the denominators count different events. The sentence to refuse has the shape *"on LoCoMo,* *system X reports [their accuracy] and this component reaches [our delivery count]"* — both halves true, the juxtaposition meaningless, and it is not written out here with real numerals precisely because it would then be quotable.

A second caution about the published numbers themselves. Several competitor scores most often quoted for this space appear in **Mem0's own comparison table** and are Mem0's reproductions rather than author-reported results; Zep's paper, for instance, reports DMR and LongMemEval and never LoCoMo. `COMPETITIVE_LANDSCAPE.md` records which numbers are author-reported and which are third-party, because the distinction survives into any comparison built on them.

This programme committed, before it had any external result, to a framing in which losing was still a finding: the question was how much of the layer survives without the generative call, not whether the deterministic version wins. **§5 reports the comparison, run here against Mem0 2.0.18 on a matched budget and a single reader.** That framing is left standing rather than quietly retired, because it is what the study was designed under and because it still governs every system in this section that was not run — which is all of them except Mem0.

One boundary is load-bearing and appears again in §13.5. The LongMemEval authors' LLM-assisted indexing and time-aware query expansion were available to this programme and were deliberately not adopted, because they add generative calls to the memory path and would change the component under test. Some of the gap between this component's external numbers and published ones is that choice, and this paper does not get to claim the choice was free.

The placement is narrow. Most systems above consume a candidate set produced upstream by similarity ranking, and this paper measures that set. The claim is not universal even across the short list: HippoRAG retrieves by graph traversal, as §2 says two paragraphs earlier, so the measurement here bears on it only where a similarity-ranked candidate set feeds the traversal.

3. The architecture

Stated first, because it is the result. What follows in §6 through §11 is the evidence for why it is this small.

3.1 What it is

Four components, and one budget.

Component	What it does
Append-only verbatim store	Every user message and its assistant reply, kept as stored. Nothing is rewritten, summarized, or distilled
Recency window	The last N episodes, unconditionally
Similarity retrieval	Cosine over embeddings against the query, admitted above a threshold
Set-level coverage objective	Chooses a delivered set by what each candidate *adds* to those already chosen, rather than scoring each independently

Everything is packed against one hard character budget, charged at exact serialized cost — per-episode tags, metadata and separators included, not just source text. That accounting was not always right, and correcting it moved published numbers by up to 68% (§12.1).

3.2 The property that matters

There are no generative model calls anywhere in the memory path. Nothing in it asks a model to write text about the store.

This is not the absence of model calls, and the distinction is one this paper's predecessor got wrong and had to correct. `context()` embeds the query on every call and `append()` embeds every episode, so an embedding model must be resident. What follows from the architecture is narrower and checkable: the memory path emits no generated text, and its output is reproducible **given a pinned embedder** — which is why the library asserts a sentinel vector hash on every store open rather than assuming one (§12.3).

Two consequences are measured rather than argued:

- **Replayability.** `context()` is a pure function of store state, query and budget, verified byte-identical across two processes. All 132 committed selection payloads and all 3 committed rendered blocks reproduce their SHA-256 byte-for-byte through the installed library, with the full suite at 1,007 tests.

- **Provenance.** Every delivered character is a stored episode verbatim. There is no generated text about the store that can be wrong, because nothing generates any.

3.3 Why those properties were bought, not designed

Interpretation, not measurement. The properties are measured; the causal story is a reading of this programme's history.

The component is reproducible because distillation was removed, and distillation was the component whose output could vary run to run. It preserves provenance because every mechanism that generated text about the store failed its own gate and was taken out. Stated plainly: **the properties that make this component deployable were bought by the failures, not by the design.** A programme that had succeeded at distillation would have shipped something harder to test and harder to audit.

3.4 One correction that travels with the description

The list in §3.1 describes the component as it now exists in the installable library, where the recency window is a genuine last-N window. **No live study behind the results in this paper ran one.** The studies ran two different rules under that name: a least-recently-delivered rotation over the whole store in the most recent work, and before it a block that locked onto the conversation's first nine turns and held them for 111 consecutive turns. Mean overlap with a true window of the same size is 0.205. §12.4 gives the measurements. Nothing in this programme establishes what a correctly implemented window would score, in either direction.

4. How results are graded

This section is what licenses the confident voice everywhere else. The programme's own operating manual is explicit that its recurring failure is a surrogate that can pass without the property it certifies, so results here are graded by how they were obtained, before anyone looks at what they say.

4.1 The standing taxonomy

Six levels, separated by **when the claim was committed relative to the number**, not by how the number was computed. Determinism is cheap; commitment order is what a result cannot buy back afterwards.

Standing	Requirement	How this paper states it
CONFIRMATORY	Pre-registered; sealed holdout; bars, endpoint and budget locked before the number existed; registration commit carries no implementation file	As an established result, with its scope cap
REGISTERED-OFFLINE	Pre-registered with bars locked first, zero generative calls, and reproducible on replay — byte-exactly where the embedding cache was retained, otherwise under recomputed embeddings, which the row says — but run on a corpus already observed, so it cannot confirm	As measured and registered, capped as characterization
REGISTERED-LIVE	Pre-registered with contrast, endpoint, budget and sample size hashed before the first generation call, but run on a corpus already observed, and **not replayable**: generation is stochastic, so replicates measure that spread instead of eliminating it	As measured and registered, capped as characterization; never as confirmation
DESCRIPTIVE	Deterministic and reproducible, but the reading was chosen after the number existed, or the quantity is a bound computed with the answer key	As measured, with what it cannot support said in the same breath
NOT DEMONSTRATED	A scored live comparison whose gap falls inside the measured 1-point band	With the number *and* the label. Not refuted either
WITHDRAWN	Corrected in ERRATA.md	Not at all. The list is paper/notes/DO_NOT_WRITE.md

The second and third levels were one level in an earlier draft, and collapsing them was a real defect: it let a byte-identical 500-store replay, a posthoc reading of an exhausted corpus, and a bound computed with the answer key all inherit the same phrase. They reproduce identically and they do not license the same sentence.

The assignment, in full, so this table is checkable without leaving the paper.

Result	Standing	The binding limit
HH-002 — 79.09% on the published harness (§5.1)	REGISTERED-LIVE	One replicate; a vendor API, so not replayable. The lead over full context was **not** the registered contrast
HH-002 — +33.31 over fixed-chunk RAG (§5.1)	REGISTERED-LIVE	The one directional claim registered before the run. Both endpoints agree
HH-002 — the RAG configuration sweep (§5.4)	DESCRIPTIVE	Built after the number existed, to account for a failed gate. Carries no claim about this component
HH-001 — head-to-head against Mem0, +7.7 points (§5.6)	REGISTERED-LIVE	This reader, this corpus, this budget, this pair of configurations. Never confirmation
NF-004 — LoCoMo pair ranking, 843→935 (§6.1)	CONFIRMATORY	Availability only; no reader, universal-rule or adoption claim
DMR-004 — no sufficiency signal (§6.2)	CONFIRMATORY	Negative; closes deterministic stopping in this arc
DMR-001 — degenerate formation (§6.3)	CONFIRMATORY	Negative; its post-stop gates are not results
DMR-001C — transfer confirmed, boundary refuted (§6.3)	CONFIRMATORY	Split verdict; the statistic was ill-chosen and is not re-scored
SAL-001 — no surprisal signal (§6.4)	CONFIRMATORY	Negative; effect ran opposite to the registration
NF-005 — turn ranking, 361→461 (§7.2)	REGISTERED-OFFLINE	Corpus already observed; capped as characterization. Does not isolate length from localization
NF-006 — statement ranking, 12→14 of 17 (§7.4)	REGISTERED-OFFLINE	One probe; art falls 2/4 → 1/4; turn 90's carrier unresolved
EC-002 — packing priority, 109→261 (§9.1)	REGISTERED-OFFLINE	Reproduction under recomputed embeddings, **not** a byte-exact replay
IC-001 — zero K episodes at 8/8 probes (§9.2)	REGISTERED-OFFLINE	Internal store; frozen candidate identities
NF-003 — three-arm, 375/388/351 (§7.3)	DESCRIPTIVE	Posthoc on an exhausted corpus. Not a registered law
AR-001 — the 5,058-character optimum (§8.1)	DESCRIPTIVE	Computed **with the answer key**. A bound, not a method
DR-002 — the pool binds structurally (§8.2)	DESCRIPTIVE	A fact about the shortlist's contents, not a measured comparison
DX-001 — the 0.0560 floor (§8.4)	DESCRIPTIVE	One frozen configuration
NF-007 — coverage floor inert (§12.6)	DESCRIPTIVE	An **instrument** stop: the test could not distinguish the arms
Study 009 memory tier, +3.0 (§13.1)	NOT DEMONSTRATED	Inside the band; process state uncontrolled
LV-001 targeted regression, −2.0 (§13.3)	NOT DEMONSTRATED	Inside the band. The kill bar firing is separate and stands
Study 011 tier isolation, −1.0 (§9.3)	NOT DEMONSTRATED	Inside the band; the correction stays rejected

Five results in the entire arc reach CONFIRMATORY, and **three of those five are negative**. That ratio is the paper's most compact structural fact: the surviving design is four components because the well-built experiments mostly returned nothing.

paper/notes/EVIDENCE_SPINE.md carries the same assignment with each number's artifact path and hash, for a reader who wants to check a figure against its source.

4.2 The machinery behind those grades

Pre-registration. Each study's design was committed before implementation, and that commit's SHA is the integrity anchor. Registration commits contain no implementation files, which is checkable and was checked. Amendments never edit a locked registration; they are standalone files recording whether they preceded the result they affect.

Preflight, in two parts, before any run. Part 1 characterizes the mechanism empirically — *not by reading the code, not by trusting its name, not by citing a prior study.* Part 2 is a ten-item checklist where "assumed" is not an answer and "verified at <SHA>" is. Two items earn their place repeatedly: PF4 asks whether the registered thresholds are achievable at all, and PF9 asks the surrogate question — *can this pass while the property it certifies is false?*

Gates that bind and run before inference. Study 008 stopped at its pre-run gates when replay proved no registered fill cap between 1 and 50 could pass the breadth and targeted gates jointly, preventing four invalid 121-turn runs. A gate is trusted to stop only after showing that its tested population and its non-stopping alternative were capable of existing; an empty join or an inert treatment is an instrument failure, not a mechanism result. §6.4 and §12.5 each report one.

Leakage control. Mechanism code — retrieval, formation, ranking, gating — may not read the answer key; measurement may. The boundary is enforced by grep, import-graph checks, and a planted test violation that must be caught.

Sealed scoring. Three blind passes with registered adjudication triggers. Every arm's scores are committed before anyone opens a mechanism log, and git order is the evidence. **The raters were three models from one family and the adjudicators were subagents — none human** (§13.8), and one live study ran a single rater against a registered three (§13.3).

None of this was uniformly achieved, and §12.6 is the list of where it was not. Four integrity gates in this programme were inert for their whole lives: three through a line-ending mismatch, and one — on the sealed holdout itself — through a constant that never matched the file it named. A targeted no-regression gate was unsatisfiable by construction for any selector. One stop was reached only because its evaluator searched for labels that did not exist. The machinery above is the standard this programme holds itself to; §12.6 is the measured shortfall against it, and the gap is the reason §12 exists rather than an embarrassment tucked into it.

4.3 What a stop means

A stop closes a design, not a question, and this paper distinguishes two kinds throughout. A **mechanism stop** means the thing was tested and did not work. An **instrument stop** means the test could not have detected the thing. Reporting the first when it was the second is how a programme accumulates false confidence, so each stop below says which it was.

5. The head-to-head: Mem0's benchmark, and Mem0 run here

This component scores 79.09% on LoCoMo. The highest row of the table Mem0 published stands at 72.90%, and this rig independently reproduced that row at 72.47% — so the comparison rests on a control it verified, not on an assumption. It gets there on 4,243 prompt tokens against the 25,405 that same row costs here.

Two studies stand behind that sentence. **HH-002** rebuilt the evaluation harness that produced arXiv:2504.19413's Table 2 — the authors' question set, answer prompt, judge prompt, judge model and metric — and ran this component through it on all 1,540 scored questions. **HH-001** installed Mem0 2.0.18 and ran it here, on one local reader at a matched budget, which is how this programme knows what Mem0's write path costs and what it loses.

Together they measure two different things and both are in this section: where the component lands against a published field, and what happens when the competing architecture is put on a bench and watched.

Provenance: notes/HH002_EVIDENCE_SPINE.md and notes/HH001_EVIDENCE_SPINE.md. Both studies hashed their commitments before the first generation call.

Scope, once, so the rest of the section can be read without footnotes. Mem0's own row was not re-run — it needs a hosted-platform account this programme does not hold — so Table 2's rows are quoted with attribution and no test in this paper is computed against them. The arm under test is NF-004's pair ranking at a 16,000-character budget, without §3's set-level coverage objective.

5.1 On the table arXiv:2504.19413 published

All 1,540 scored LoCoMo questions — every question in the ten conversations except the adversarial category the harness itself skips.

System	LLM-as-a-Judge	Mean prompt tokens	Source
This component	**79.09%**	**4,243**	measured here

System	LLM-as-a-Judge	Mean prompt tokens	Source
Full context	72.90%	—	Table 2
Full context, reproduced here	*72.47%*	*25,405*	*measured here*
This component, undated turns	71.56%	3,696	measured here
Mem0^g	68.44%	—	Table 2, Mem0's own system
Mem0	66.88%	—	Table 2, Mem0's own system
Zep	65.99%	—	Table 2, run by Mem0, not by Zep
LangMem	58.10%	—	Table 2, run by Mem0, not by LangChain
RAG, best variant	60.53%	—	Table 2
OpenAI memory	52.90%	—	Table 2
A-MEM	48.38%	—	Table 2, run by Mem0, not by A-MEM
No memory	26.30%	84	measured here

Five of the six quoted rows are the Mem0 authors' reproductions of other people's systems. Zep's own paper reports DMR and LongMemEval and never LoCoMo. Figure 1 draws the table, with the rows this rig measured separated from the rows it quotes and the floor of §5.3 drawn across all of them.

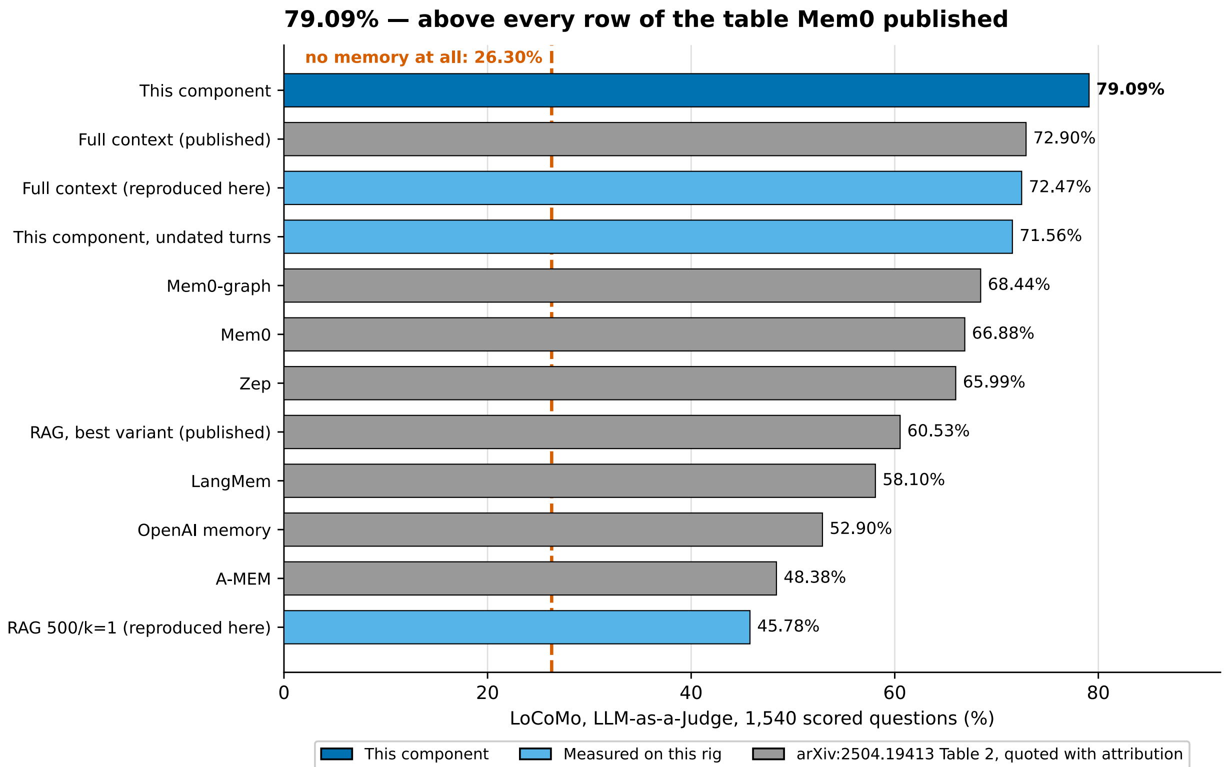

Figure 1. Where this component lands on the table arXiv:2504.19413 published. *Every row scored on the same 1,540 LoCoMo questions, the same harness, the same judge.* Dark blue is this component at **79.09%**. Light blue are the rows this rig measured, including full context reproduced at **72.47%** against the published **72.90%** — the control that licenses printing the rest together. Grey rows are quoted from Table 2 and were not re-run; no test in this paper is computed against them. The dashed line is the study's second finding: an empty context scores **26.30%**, and because the judge prompt, model and question set are shared, that floor sits under every bar in the chart. Sources: A_CDW/judged_r1.json 6f56cebcdd34fdfb, A_FULL/judged_r1.json 31933075d01b24e3, A_NONE/judged_r1.json 4b12162984f80777, commitments.json 1ca3931c8b97dd7b.

The component's row sits above every quoted row, and it does so at a sixth of the prompt tokens of the arm that previously topped the table. Against full context **reproduced on this rig** it leads by **6.62 points** — 210 gains against 108, one-sided exact binomial p = 5.593e-09 — and the deterministic endpoint, which involves no model, agrees at **+15.32 points**, p = 9.309e-31.

That contrast is post-hoc, and the registration got its sign wrong. HH-002 registered exactly one directional claim, and predicted this component would land *between* 60.53% and 72.90% — below full context. It did not. A p-value on a direction that was not registered, in a comparison whose sign the registration mispredicted, is reported for completeness and carries no confirmatory weight.

The registered claim is the one that cleared its bar by the widest margin. Against fixed-chunk RAG — the contrast written into `commitments.json` before the first generation call — this component gains **33.31 points: 558 items won against 45 lost, one-sided exact binomial p = 6.615e-114.** The deterministic endpoint agrees at **+24.16 points**, which is the condition the registration set in advance for making any directional claim at all. Both endpoints, one of them model-free, point the same way on the one comparison this study committed to before it had a number.

5.2 The rig reproduces their ceiling row to within half a point

Before measuring itself, this rig measured them.

Full context is the row that cannot be mis-specified: `--chunk_size -1` hands the model the whole conversation, with no chunking, no embedder, no retrieval and nothing to configure. **This rig scored it at 72.47% against a published 72.90% — a gap of 0.43 points**, against a tolerance of ±3.0 fixed before any number existed.

Half a point on a 1,540-question benchmark, reproducing a figure published by a different team on different hardware. On the one row where nothing can be misconfigured, the corpus adaptation, the prompt reconstruction, the judge and the metric all land where theirs did. That is what puts this component's number on the same axis as the rest of the table — placement, and nothing more.

The judge is not the noise source either. The same 1,540 sealed answers, scored twice, landed **0.06 points apart** — 3 items moved out of 1,540.

5.3 LoCoMo has a 26-point floor that its own paper never reports

Run the benchmark with **no memory block at all** — an empty context, the model guessing — and it scores **26.30%**.

G-FLOOR failed. The registered bar was below 5%, on the expectation of near-zero, and 26.30% is not near zero.

It is also this study's second finding, and not the one it went looking for. Guessability alone accounts for it, without any need to suppose the model has seen the corpus: the failure is a judge instructed to be generous meeting questions a reasonable guess can satisfy. No contamination probe was run against this reader, so contamination is unmeasured here rather than excluded. The judge prompt, reproduced byte-exact from the upstream blob, tells the grader to count an answer correct if it touches the same topic as the gold answer. `Last Saturday` scores CORRECT against `The weekend before 22 July 2023`. `family members` scores CORRECT against `Family`.

That prompt, that model and that question set produced every row of Table 2, so the floor belongs to the instrument — the reader, the grader and the questions — rather than to any system standing on it, and **it sits under all of them**. That last step is an inference from shared instrumentation, not an observation: those runs were not watched.

The floor is not a property of the corpus either. HH-001 ran the same empty context on a local 27B reader and scored **0.000** (§5.6). Change the reader or the grader and the floor moves; LoCoMo did not become guessable, this pairing of model and judge made it so.

The floor is not uniform either. It is highest on **multi-hop at 38.54%** — the questions meant to be hardest — and **32.34% on open-domain**, which is 841 of the 1,540 questions and the largest stratum by far. It is lowest on temporal, at **11.21%**. A benchmark whose biggest category is a third answerable by guessing, and whose multi-hop questions are more guessable still, measures less than its headline suggests, and every system quoted on it inherits that.

Measured from the floor, the rows this rig produced spread out:

Arm	Raw	Above the 26.30% floor
This component	79.09%	**52.79**
Full context, reproduced here	72.47%	46.17
Fixed-chunk RAG, reproduced here	45.78%	19.48

Only rows measured on this rig appear in that column. The floor was measured here and varies by stratum; subtracting it from a row whose strata were never published would be arithmetic wearing the clothes of a measurement.

5.4 One RAG setting is worth 26 points — more than the table's whole spread

The second registered reproduction target was fixed-chunk RAG at 500 tokens and one chunk, mapped to Table 2's 60.53%. It scored **45.78%. G-CTRL failed**, by 14.75 points, and the registration had committed in advance that a failure is the result.

Chasing it produced the study's third finding. Table 2's row is labelled *RAG (best variant)* — the top of a sweep the paper never specifies — so the registration had bound a gate to one recipe out of a family. Running the family:

Variant	Mean prompt tokens	Score
500 tokens, 4 chunks	2,030	**65.32%**
1000 tokens, 2 chunks	2,012	50.65%
500 tokens, 1 chunk — the registered target	570	45.78%
1000 tokens, 1 chunk	1,047	39.16%

The sweep spans 26 points — wider than the gap between the top and bottom halves of Table 2 — and the published 60.53% sits inside it. A row that moves that far on a configuration choice names a family, not a number.

The explanation does not cancel the failure. The gate was registered against a value, it missed by 14.75 points, and the sweep that accounts for it was built after the number existed and is DESCRIPTIVE. The registered arms are not.

Two results came out of it. The component leads the best variant the sweep found by **13.77 points**. And at a **matched budget** — 2,030 prompt tokens against 2,012, inside 1% — four 500-token chunks beat two 1000-token chunks by **14.68 points**, 318 gains against 92. **The same budget spent on more and smaller units is worth fourteen points.** Size and count move together here, so this does not isolate which one carries it; it points the same way as §7 through a mechanism §7 never tested, and inherits none of §7's standing.

5.5 What the dates bought

The harness renders every turn with its session timestamp. NF-004's candidate unit carries no timestamp, because NF-004's endpoint was whether evidence text was delivered and no date was needed to answer that. HH-002 ran both renderings, registered in advance, and predicted the gap would sit in the temporal category.

Category	n	Dated	Undated	Effect
Temporal	321	68.54%	32.09%	**+36.45**
Single-hop	282	71.63%	72.34%	−0.71
Multi-hop	96	55.21%	57.29%	−2.08
Open-domain	841	88.35%	87.99%	+0.36
Overall	**1,540**	**79.09%**	**71.56%**	**+7.53**

The whole overall gap is one stratum, and Figure 2 shows it. The other three move by 0.71, 2.08 and 0.36 points — two of them in the undated arm's favour, none of them in a direction the ablation predicted. A retrieval unit that drops the timestamp is not slightly worse across the board; it is catastrophic on one question type and indistinguishable on the rest.

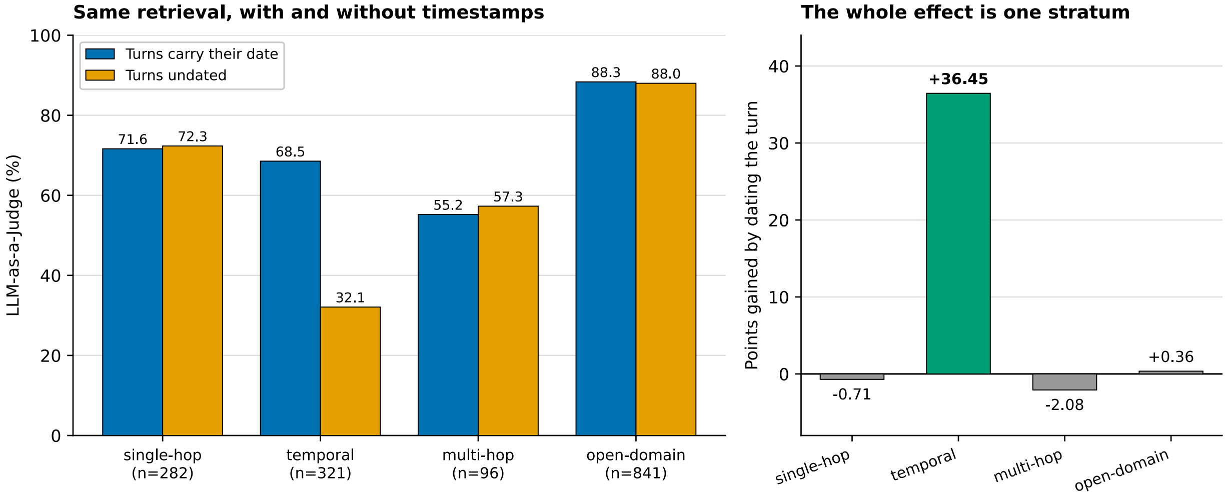

Figure 2. What the timestamp buys, by question category. *The same retrieval, run twice, differing only in whether a delivered turn carries its session date.* Registered in advance with the prediction that the gap would sit in the temporal category. It does, and nowhere else: **+36.45 points on temporal**, against −0.71, −2.08 and +0.36 on the other three. The overall 7.53-point difference is one stratum of 321 questions; a retrieval unit that drops the timestamp is not uniformly worse, it is disabled on one question type and unchanged on the rest. Sources: A_CDW/judged_r1.json 6f56cebcdd34fdfb, A_CDW_NOTS/judged_r1.json cbedd6689e2c6d9b.

The same table answers a question about the ceiling arm. Full context scores **49.53%** on temporal questions against this component's 68.54%, despite holding every turn and every date in its window. Having the evidence present is not the same as having it findable.

5.6 Mem0, run here

HH-001 is the study that watched Mem0 work. Five memory layers, 300 questions drawn by seeded stratification from 850 answerable LoCoMo records, three replicates each, one reader, one judge, one 16,000-character budget.

Arm	Memory layer	Judged	Containment
A1	Whole conversation, unbudgeted	0.613	0.320
A2	**This component**	**0.563**	**0.313**
A4	Fixed-width chunk retrieval	0.550	0.287
A3	**Mem0 2.0.18**	0.487	0.217
A0	No memory	0.000	0.000

Figure 3 sets the two panels side by side. **The component answers 7.7 points more of the same questions than Mem0 does.** Paired over the 300 items: **46 gains, 23 losses**, one-sided exact binomial **p = 0.0038**. The deterministic containment endpoint puts the same contrast at **+9.7 points, 40 gains against 11, p = 2.85e-05**. Both endpoints point the same way, which is the condition this study registered in advance for making a directional claim at all.

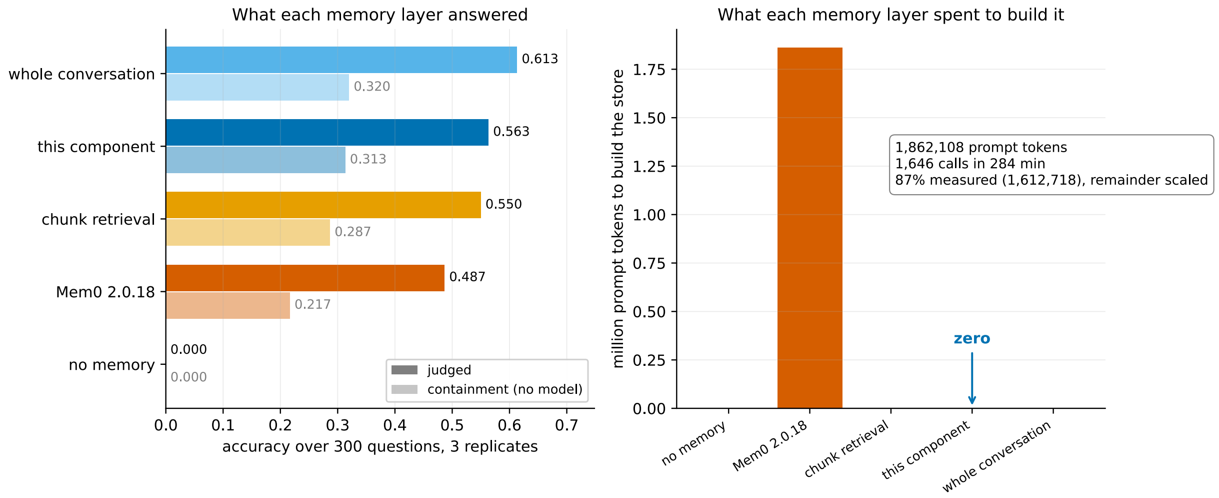

Figure 3. The head-to-head. *What each memory layer answered, beside what it spent to build the store.* Judged and containment accuracy over 300 questions at three replicates, one local reader, one 16,000-character budget: whole conversation 0.613, this component 0.563, fixed-width chunk retrieval 0.550, Mem0 2.0.18 0.487, no memory 0.000. The right panel is **prompt tokens spent building the store — 1,862,108 for Mem0 against zero for every other arm**, across 1,646 calls and 284 minutes. That total is RECOMPUTED: the counter overlaps 86.6% of the ingest and the overlap is scaled by wall clock, over a MEASURED floor of 1,612,718. It supersedes a published 5,988,818 that had booked the reader phase to the ingest (Amendment 001). The two panels are not one trade curve: accuracy and build cost are different quantities and a single axis would imply a relationship the data does not describe. Sources: result.json 7fa4119c29f06b1c, cost/mem0_ingest.json a9653199d0d8317f, cost/storage.json adcc2ea410046e22.

§13.1's 3.0-point band does not govern this contrast. That band was measured on a 13-point holistic rubric scored once per run. This is 300 items paired within item, three replicates deep, read out as a discordant count — a different instrument with its own noise reading, and this one reports it: per-item unanimity across replicates runs 0.85 to 0.89 by arm.

The floor arm scored **zero** here. That is the one number HH-002 overturned: on a local 27B reader an empty context answered nothing, and on GPT-4o-mini with the vendor harness's generous judge the same empty context answers 26.30%. The floor is a property of the reader and the grader, not of the corpus.

5.7 What the generative write path cost and lost

Mem0 built its store with **1,646 generative calls over 284 minutes**, one call for every message pair, carrying **1,131 prompt tokens each — 1,862,108 across the ingest**, and writing back nearly as many again. This component built its store with **none**. That zero is architectural rather than measured: append() embeds and stores, and no code path asks a model to write text about what was stored.

That token figure is a correction, and it cuts against this paper. It was published as 5,988,818. The counter it came from is cumulative and process-wide, and it ran 105 minutes past Mem0's last write; 73% of the total was the reader phase on the same server. The corrected value is 3.22× smaller, and the claim that per-pair cost climbs as the store grows fell with it — inside the clean overlap the rate is flat to declining, at −0.444 tokens/min per stored memory. What does drift is latency, 1.13× first to last decile, which is what Figure 4 shows and always showed. Amendment 001 carries the derivation.

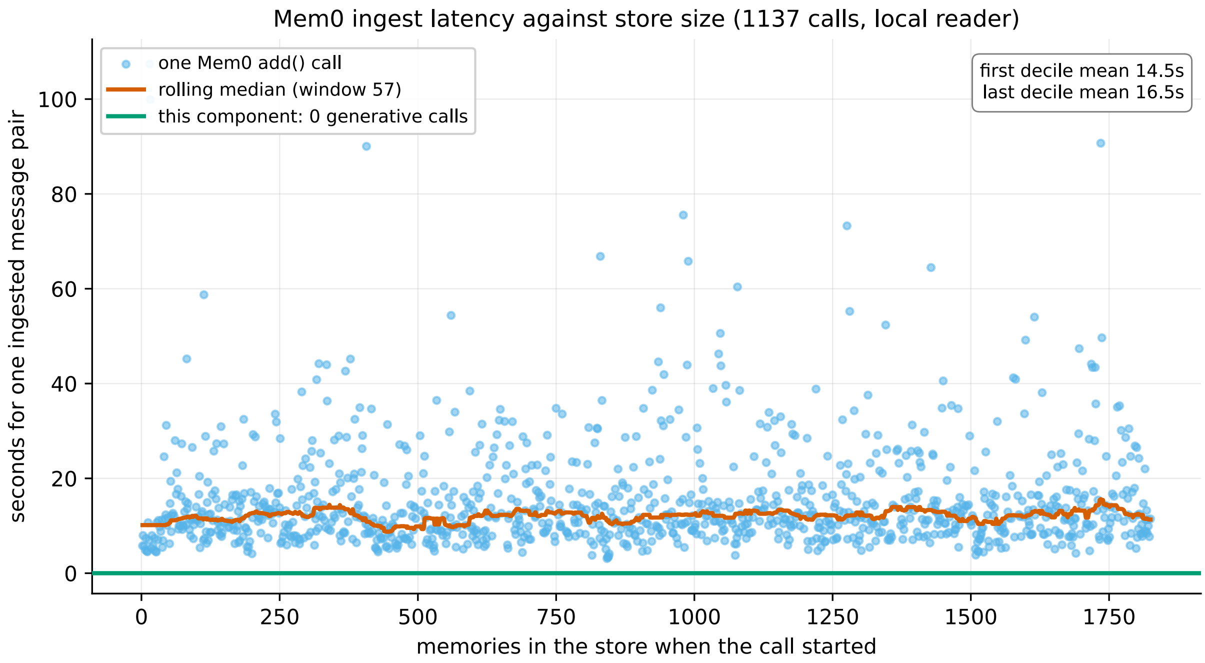

Figure 4. What the store costs to build, as it grows. *Mem0's per-pair ingest latency against the number of memories already stored.* Derived from Mem0's own history table: each add() emits a burst of writes milliseconds apart, and the gap between bursts is the wall clock that call spent. 1,137 measured bursts, store reaching 1,825 memories, p50 11.9 s and p95 34.8 s. First decile mean 14.5 s against last decile 16.5 s — a 1.13× drift, not a runaway. The flat line at zero is this component's generative cost, which is architectural rather than measured. **The latency points cover only the 69% of pairs that wrote something**; a pair Mem0 declined to store leaves no burst and is invisible here, and is likely the faster case. Source: cost/mem0_ingest_latency.json 0ad65363b635b6e7.

The comparison does not depend on the size of that number. 1,646 calls against zero is the architecture, and a call count is the honest unit for it.

Of the 315 answers stated verbatim somewhere in the six source conversations, **66 are absent from Mem0's finished store — 21%.** This is a containment test over the store's own memory text: a preserved paraphrase counts as absent, so **21% is an upper bound on extraction loss and cannot understate it**. That is the honest reading and it is the one to quote.

Two further numbers come from the ingest itself. **509 of 1,646 message pairs — 31% — produced no memory at all**: the extractor received the turn, spent a model call on it, and decided nothing in it was worth storing. And **16 extractions returned malformed JSON** and were discarded, 0.97% of the corpus, logged by Mem0 and passed over in silence.

Downstream, the answer reaches the delivered block for **101 of 108** eligible items under this component and **79 of 108** under Mem0 — a 0.935 survival rate against 0.732. A verbatim store cannot do better than its selector, and cannot do worse than what it stored, because it stored the turn unchanged.

Retrieval separates them again. Assembling one context block takes this component **10 milliseconds** at the median and Mem0 **413 — 41 times slower**, on the same machine. The finished store occupies **42.8 MB** against **7.2 MB**, counting only Mem0's vector store and excluding its 692,224-byte history log.

The cost runs the other way at read, and the paper states it in the same breath. Mem0 is the cheapest memory arm per question — **3,392 prompt tokens against this component's 4,009** — because model-extracted memories are shorter than stored turns. A system written once and queried a million times amortises its ingest away. Which architecture is cheaper is decided by a read-to-write ratio neither paper states.

5.8 What the other arms did better

Fixed-width chunk retrieval stores **2.8 MB against this component's 7.2** and reads at **3,904 prompt tokens against 4,009**. It is smaller and cheaper on both counts, and it scored 0.550 to this component's 0.563 — thirteen thousandths. This component also has the **lowest replicate agreement of any memory**

arm in HH-001 — 0.853 against 0.870 and 0.891 — so its answers are the least stable of the three under reseeding.

Splitting HH-001 by how far back the evidence sits — these are 369- to 680-turn transcripts — the component leads Mem0 by **11.9 points in the oldest quarter** and **14.9 points in the newest**, and trails by 3.1 in the second quarter. The advantage is not a recency effect and is not uniform.

5.9 The three boundaries that bind this section

Stated once, in full, and not repeated. §13 carries the rest.

Six of the eleven rows in §5.1 were not run here. They are quoted from Table 2 with attribution, no test in this paper is computed against them, and the component is placed beside them rather than measured against them. The one competing system this programme did run is Mem0 2.0.18, in §5.6.

The lead over full context was not registered. §5.1 states it and labels it. The registered contrast is the component against fixed-chunk RAG, at +33.31 points.

Both studies are `REGISTERED-LIVE`, not `CONFIRMATORY`. LoCoMo is exhausted on both splits and generation is stochastic, so neither can confirm and neither becomes confirmation by being re-described. §6 holds the results that can.

Scope on capacity and breadth: LoCoMo fits a modern context window, so this benchmark cannot test reach, only cost and accuracy — and both arms carry NF-004's pair ranking without §8's set-level coverage objective, so neither tests breadth.

6. The confirmatory results

Five results in this arc were obtained under a sealed holdout with bars locked before the number existed. This section reports all five. Three returned nothing, and they are here at the same resolution as the two that did, because a programme that reports only its sealed successes has not earned the standing its sealed successes carry.

6.1 NF-004 — ranking granularity, confirmed on withheld LoCoMo conversations

Standing: CONFIRMATORY. Pre-registration `95f0d25c`. Ten LoCoMo conversations were split whole-conversation — so no question could share dialogue across the split — by a seeded hash committed before any question, answer, category or evidence list was opened. Four went to development. **Six were sealed and not touched until the bars, the endpoint and the budget were locked.**

The question is what a candidate should be scored *by*. The baseline gives every adjacent-turn pair the maximum query cosine attained by any pair in its session — a session-level score inherited downward, which is what a system does when it ranks sessions and then packs their contents. The treatment gives each pair **its own** cosine. Both pack identical candidates under an identical 16,000-character skip-on-overflow rule. Nothing else differs.

On 1,098 fully resolvable question-answer records, with **zero model calls and zero embedding calls during measurement**:

Arm	Complete evidence delivered	Any evidence
Source order (no ranking at all)	258 / 1,098	352 / 1,098
S_SESSION_RANK — inherited session score	843 / 1,098	950 / 1,098
`P_PAIR_RANK` — own cosine	**935 / 1,098**	**1,027 / 1,098**

140 gains, 48 losses, 910 ties. Net +92. Gain/loss ratio **2.92** against a registered bar of 2.0. One-sided exact binomial **$p = 6.19e\text{-}12$** over the 188 discordant questions.

That p assumes the questions are independent, and they are not — they cluster within six conversations. The registered statistic is not re-scored, but a conservative alternative is reported beside it: treating each conversation as a single observation, all six are net positive, one-sided sign test **$p = 0.0156$**. Six observations is what the sealed design actually bought, and it still clears 0.05. Median packed characters 15,986 against 15,988, so the treatment is not simply spending more. Median best-evidence rank moves **9 to 2**; p90 moves **80 to 34**, and those rank statistics were opened only after the disposition was committed.

Every one of the six sealed conversations is net positive: +7, +6, +13, +30, +15, +21. So are all five source categories. At the 32,000-character secondary the direction holds, 961 to 1,024.

Gates G0 through G7 all pass, including a vector seal reading 2,749 of 2,749 cached vectors with zero misses, and a G7 replay whose SHA-256 equals the committed G6 hash. Figure 5.

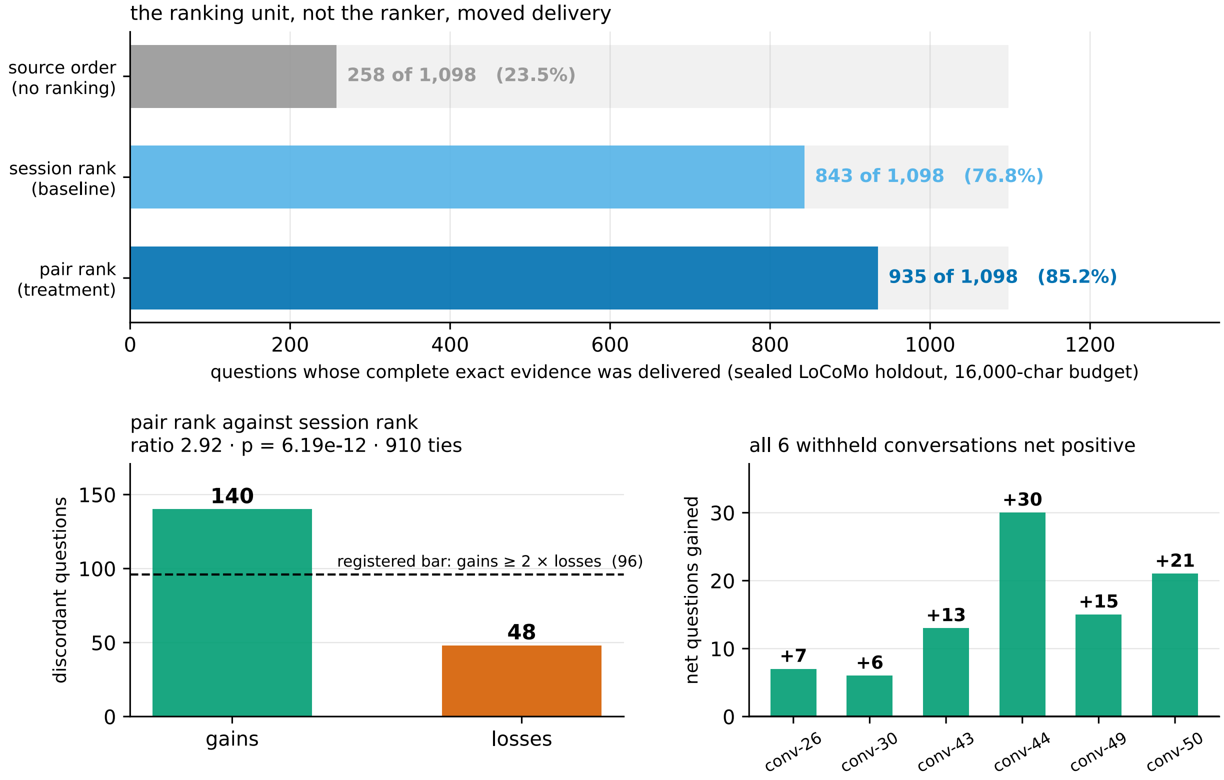

Figure 5. The sealed holdout. *Six LoCoMo conversations withheld until the bars were locked, and the ranking unit is the whole difference.* Complete exact-evidence delivery over 1,098 records at a 16,000-character budget: source order 258, session-score inheritance 843, own-cosine pair ranking 935. The gain/loss panel shows 140 against 48 with 910 ties, against the registered bar of gains at least twice losses — ratio 2.92, one-sided exact binomial p = 6.19e-12. Every one of the six withheld conversations is net positive: +7, +6, +13, +30, +15, +21. Median packed characters differ by 2, so the treatment is not spending more. **The result is bounded to availability and authorizes no reader claim.** Sources: g6_holdout_outcomes.json 7be2668d21163c93, g7_result_integrity.json 890b4831d530e9a7, NF_004_PRE_REGISTRATION.md de2d5e05646b769c.

Scope cap, and it is binding. This is availability: whether the text carrying an answer was present in the delivered context. It is not accuracy, and the registration authorizes **no reader, live, universal-rule, promotion or adoption claim**. The universal-rule term is the one that governs §7 and §14: this confirms one substitution on one corpus, not a law about granularity. §13.3 is where the accuracy distinction has teeth, and §7.3 is where the universality one does.

What the source-order control buys. 258 against 843 is the reason the treatment effect cannot be read as a budget artifact. If the budget were slack enough that ordering barely mattered, the unranked control would sit near the ranked arms. It sits 585 items below. A development-side sweep across budgets from 4,000 to 96,000 characters makes the same point at every truncated point, and ties only at 96,000, where everything fits and ordering is irrelevant by construction.

6.2 DMR-004 — no mechanical sufficiency signal *(negative)*

Standing: CONFIRMATORY. The strongest confirmatory construction in the repository, and it returned nothing.

The question: can a model-free precedence parser over query text alone identify when a query's evidence obligations are *finite* — when a retriever can know it has enough and stop? A positive result would have given the deterministic stack an adaptive controller. Sealed 180-query holdout, two blind raters — neither human, see §13.8 — and PF3 verifying the entire commit ordering from git history: protocol, then labels, then registration, then compiler, then holdout labels, then gates, with the registration commit carrying exactly one file.

Registered criterion	Bar	Result
Youden's J	≥ 0.50	**0.320 — FAIL**

Registered criterion	Bar	Result
False-finite rate	≤ 0.15	**0.188 — FAIL**
LOOKUP recall	≥ 0.60	0.800 — PASS
Well-formed span share	—	1.000 — PASS

Raw accuracy was 0.706. An always-OPEN degenerate control scores **0.650**, which is precisely why J and not accuracy was the registered statistic: a 5.6-point margin over answering "I cannot tell" to everything is not a controller. The two raters — the implementing agent and the carried local model — reached J of about 0.76 against the compiler's 0.320, with Cohen's kappa of 0.770, so the task is doable and the mechanism did not do it. The registration records the boundary before any label existed: rater A is not independent of the mechanism, and **neither rater is human** (§13.8).

The failure structure is specific. Of 31 misses, 12 are questions of the form *which happened first* — a family flagged in writing before the compiler existed and deliberately not patched, because patching a known family after seeing its cost is the rescue AGENTS.md §9.4 forbids. Adding the registered markers first and last moved development J from 0.363 to **0.220**: implementing the lock faithfully made it worse.

Disposition: a model-free adaptive controller is not authorized on this evidence, and the registration forbids replacing the compiler with a second language-model call inside this arc. That closes the deterministic stopping line, which is why the shipped component has no adaptive controller in it.

6.3 DMR-001 and DMR-001C — event formation *(negative, then split)*

DMR-001. Standing: CONFIRMATORY. Does an absolute embedding-drift threshold form a usable event substrate? 2,000-episode sealed holdout. **52 of 74 events close because the size cap binds — forced fraction 0.703 against a bar of 0.35.** The size cap became the partitioner. Drift precision was perfect on the boundaries it did find, 20 of 20, and irrelevant, because 0 of 52 forced boundaries matched anything. The locked threshold sits above the holdout's 95th percentile while firing on 18.5% of development episodes against 1.2% of holdout — an absolute drift threshold is not a transferable quantity.

Two cautions travel with this. The second gate check was **unreachable by construction**, a preflight defect recorded rather than repaired; the disposition does not depend on it. And the two descriptive gates were computed after the stop — they are not cited as results anywhere in this paper.

DMR-001C. Standing: CONFIRMATORY, and it splits. A relative percentile rule was frozen at its predecessor's anchor, and 50 unread LongMemEval haystacks — 11,453 episodes, 2,128 real session seams — were fetched *after* the freeze.

- **Transfer confirmed.** Per-stream fire rate holds between 3.41% and 7.35%, p05 to p95 ratio **1.67x**, against the fixed rule's 9x to unbounded. The operating point moves across corpora; the relative rule survives the move.
- **Boundary claim refuted.** Macro F1 of **0.387** against a control that chops every four episodes regardless of content, at **0.606**. The detector loses to a fixed chop by 0.219.

The report then audits its own statistic: macro F1 against a corpus with seams every 5.4 episodes rewards frequent firing, so it was a poorly chosen bar. **It is not re-scored.** Choosing a better statistic after seeing the number is the same rescue the previous section refused.

6.4 SAL-001 — no independent surprisal signal *(negative)*

Standing: CONFIRMATORY. Does a surprisal-proximity signal identify what deserves capture, independent of what is already retrievable? Deterministic 60-history LongMemEval holdout, label-blind scoring sealed, 92 session-level AUC replications.

Adjusted neighbour AUC **0.41599** against a bar of 0.60. One-sided permutation **p = 0.99134** against a bar of 0.01. Bootstrap 95% interval [0.35132, 0.48388]. **Five of six strata fall below chance.** The registered effect is not weak; it is in the opposite direction. That kills the surprisal-based capture line the programme's design document had proposed.

6.5 What five sealed experiments bought

One positive result, on the unit of ranking, that reproduces at the turn and pair unit on all three corpora and on every conversation inside the sealed one, with §7.3 marking where the direction reverses at a coarser unit. Three well-built negatives that each closed a line the programme wanted: adaptive stopping, event segmentation, and surprisal-based capture. One split verdict where the transfer property confirmed and the claim built on top of it did not.

The shipped component contains none of the three killed mechanisms. That is the subtraction §11 completes, and this section is where most of it was paid for.

7. Granularity: rank at the finest informative unit

§6.1 confirmed one substitution on withheld data. This section locates the mechanism using corpora already observed — reported as measured rather than confirmed — and §7.3 reports the case where the same lever, applied at a coarser unit, **reverses sign**. The reversal is on the same corpus as the largest gain, which makes it the more informative half of this section.

7.1 The size of a candidate

Standing: REGISTERED-OFFLINE for NF-005 and NF-006; DESCRIPTIVE for NF-003's three-arm reading in §7.3. Zero model calls throughout.

Unit	Median characters
LongMemEval evidence *episode*	**2,550**
The exact source *turn* carrying the answer	**298**
LoCoMo adjacent-turn pair	**241**

An embedding of a 2,550-character episode is an average over everything that episode discusses. If one sentence in it answers the query, the query's match against the whole is diluted by the rest. This is measurable rather than rhetorical: across the corpus, **longer parents have worse normalized own-cosine evidence rank at Spearman rho 0.484**. Separately, 831 of 881 evidence flags fall on user turns rather than assistant turns, so the dilution is asymmetric in a way that matters for what a system should index.

7.2 NF-005 — splitting episodes into their source turns

465 turn-labelled LongMemEval questions, 32,000-character budget, packing unit held fixed at the turn so that only the *ranking* unit varies.

Ranking unit	Packing unit	Any exact evidence	All exact evidence
Source order	Turn	64 / 465	7 / 465
Episode	Episode	351 / 465	201 / 465
Episode	Turn	361 / 465	208 / 465
Turn (own cosine)	Turn	**461 / 465**	**454 / 465**

100 gains, 0 losses, 365 ties. One-sided exact binomial p = 7.89e-31. Median best evidence rank moves 5 to 1; p90 moves 131 to 7. Figure 6 places this beside the other two corpora and the candidate-size measurement that explains all three. The budget delivers 109 turns where it delivered 46 episodes.

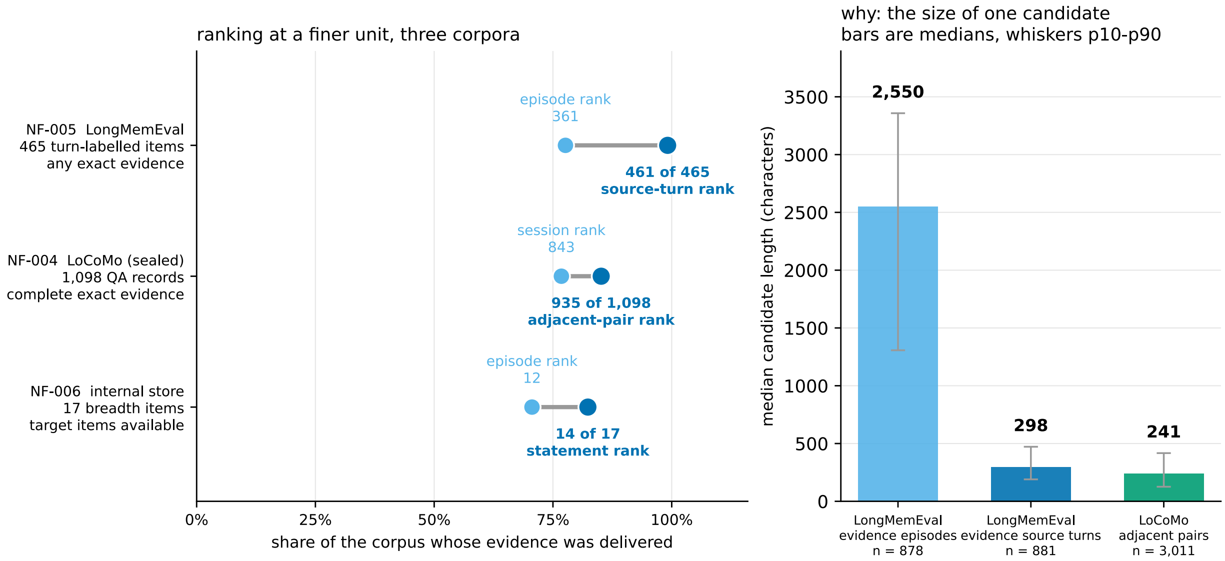

Figure 6. Granularity across three corpora. *The same substitution, three corpora, and the candidate size that explains it.* Left: exact-evidence delivery before and after ranking each candidate by its own cosine — LongMemEval 361 to 461 of 465, LoCoMo 843 to 935 of 1,098, the internal store 12 to 14 of 17. Right: median candidate length with p10–p90 whiskers — LongMemEval evidence episodes 2,550 characters, their exact source turns 298, LoCoMo adjacent pairs 241. An embedding of the 2,550-character parent averages away the 298-character turn that answers the question. LongMemEval's gain/loss ratio is undefined rather than large — 100 gains and zero losses — and is drawn off-scale with its raw counts. §7.3 marks where the direction reverses at a coarser unit. Sources: g8_integrity.json 17cbdbad274c1863, exploration.json 00d78ad3bd113abb, g6_holdout_outcomes.json 7be2668d21163c93, g8_g9_measurement.json d20dffd563e5777e.

Gates G0 through G8 pass. The vector seal reads 167,918 cached vectors with zero misses. The final outcome hash and its replay hash are identical.

Scope cap. Splitting an episode changes its character count *and* its semantic localization at the same time. This result supports candidate information dilution as the moderator; **it does not isolate raw character count**, and no experiment here does.

7.3 NF-003 — where the rule reverses, and why it is stated conditionally

The same corpus, varying ranking and packing independently:

Ranking unit	Packing unit	Exact answer-episode delivery
Session	Session	375 / 465
Session	**Episode**	**388 / 465**
Episode	Episode	351 / 465

Finer *packing* gains 13 items. Finer *ranking* — at this unit, on this corpus — **loses 37**, with 26 gains against 63 losses. That is the opposite sign to §7.2, on the same corpus, and the reconciliation is the size table in §7.1: an episode is already small enough that its embedding stays informative, and dropping to it discards the broader context that was doing the scoring work. A source turn is small enough that its embedding is *sharp*. The 63 items rescued by coarse ranking have median own-episode cosine rank 46; the 26 gained by fine ranking have median rank 10.

State the shape before stating any rule, because the shape is not monotone. On one corpus, at a fixed packing unit, the episode loses in both directions: the turn beats it 461 to 361, and the session beats it 388 to 351. A candidate unit that is worse than both its parent and its child cannot be produced by any rule of the form *finer is better* or *coarser is better*. Whatever governs this is not a monotone function of granularity, and this programme has not identified it.

What can be said conditionally is: **rank at a unit small enough that its embedding is dominated by the material answering the query, and pack at the finest affordable unit.** The first clause needs an operational test to be a rule rather than a restatement, and **it does not have one here** — "dominated by" is measured after the fact by whether delivery improved. Two proxies are available and neither was registered: the 240-to-300-character band where every confirmed gain in this paper sits, and the Spearman rho of 0.484 between parent length and worse own-cosine rank. Both are descriptions of these corpora, not thresholds anyone should carry.

This is a posthoc characterization on an exhausted corpus, not a registered universal law, and §7.5 refutes the scope condition that was proposed to predict where the sign flips.

7.4 NF-006 — the same substitution on the internal store

The internal 121-turn store, 119 eligible episodes, one enumeration probe worth 17 items, 32,000 serialized characters. Selections were sealed outcome-blind before measurement.

Arm	Total	Civil	Art	Monetary	Marine	Units	Chars
Episode rank, episode pack	12/17	5/5	2/4	1/4	4/4	15	31,569
Inherited score, statement pack	7/17	3/5	2/4	1/4	1/4	51	31,931
Own-statement cosine, statement pack	**14/17**	5/5	1/4	**4/4**	4/4	80	31,991

The middle row is the control that makes this a ranking result rather than a packing result: split the candidates but let each inherit its parent's score and delivery *falls* to 7 of 17. The unit only helps when it is scored on its own terms.

Targeted probes hold at 21 of 21 against 21 of 21 — zero losses. Gates G0 through G9 pass.

Two boundaries travel with this number. The winning trace selects **no statement whose source turn is 90**, so the specific episode a prior diagnostic identified as the hardest miss remains unresolved; all four monetary items arrive by another route. And art falls from 2 of 4 to 1 of 4. This is a breadth composition trade, not universal dominance.

7.5 Where a tempting generalization fails

A budget sweep across both external corpora refutes the obvious scope condition. The proposal was that the effect depends on how oversubscribed the budget is — the ratio of available candidate text to deliverable characters. **Seven overlapping cells have opposite signs.** The sharpest pair: LoCoMo at a 4,000-character budget, median binding ratio 19.85x, nets **+123**; LongMemEval at 24,000 characters, median ratio 19.39x, nets **−14**. Nearly identical pressure, opposite direction.

The binding-ratio scope condition is therefore rejected, and the registration that followed was written corpus-specific rather than universal. The 16,000-character operating point in §6.1 was chosen because its baseline sits off-ceiling at 80.9% of deliverable items — **explicitly not because it showed the largest effect.**

8. The decomposition: three constraints in a forced order

The granularity result is about the unit. This section is about what happens after the unit is fixed, and it is the part of the programme that generalizes least and explains most. All of it is **DESCRIPTIVE** on the internal store: the optimum is computed with the answer key, the pool claim is a fact about the shortlist's contents rather than a comparison, and the floor is one frozen configuration.

8.1 Capacity was never the constraint

The measurement that makes the rest possible: compute, with the answer key, the cheapest set of stored episodes that satisfies the enumeration probe, charged at the same exact serialized cost as the deployed path.

Configuration	Items available	Characters spent
Deployed baseline	6 / 17	31,946
Shipped set-level configuration	12 / 17	31,569
Exact known optimum	**14 / 17**	**5,058**
Greedy variant	15 / 17	5,455

A perfect picker reaches the target using a sixth of the budget. Deployed selection spent all of it and delivered a third as much. Figure 10.

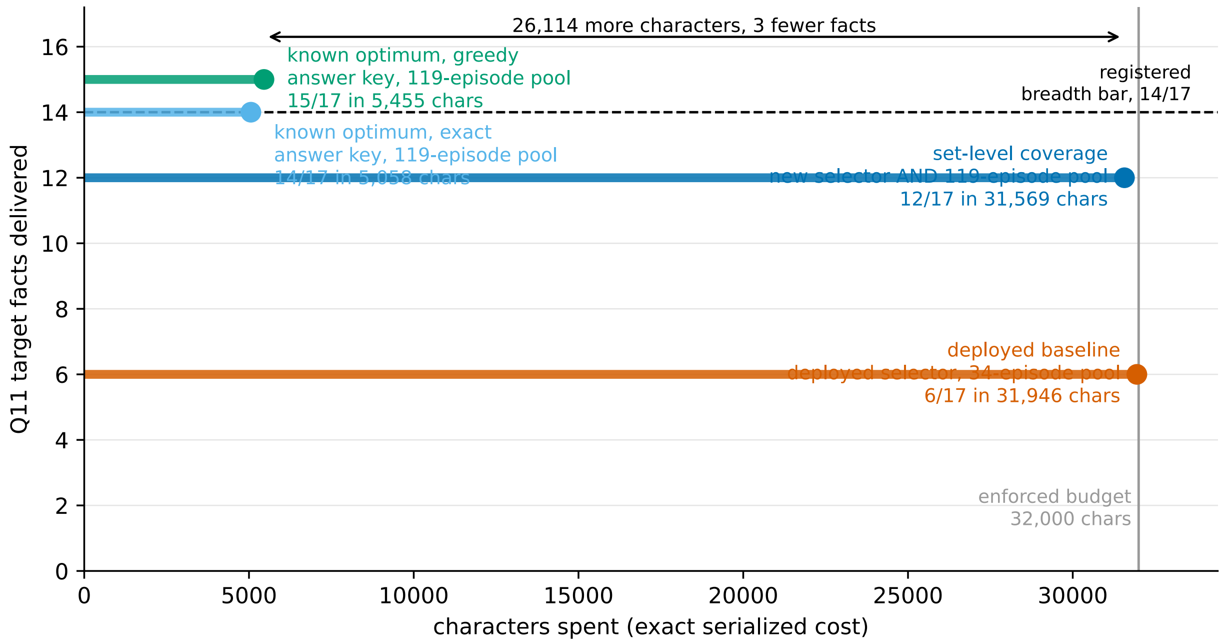

Figure 10. The budget efficiency gap. *The constraint was never capacity: the tallest result is also the narrowest.* Each horizontal stem runs from zero to the characters spent, at a height equal to the items made available, over the same store at the same enforced 32,000-character budget. The deployed baseline reaches 6 of 17 for 31,946 characters; the shipped set-level configuration 12 of 17 for 31,569; the exact known optimum 14 of 17 for 5,058, leaving 26,942 unused; its greedy variant 15 of 17 for 5,455. **The top two bars change two things against the baseline, not one** — the selector and the candidate pool; §8.3 separates them, and on the deployed pool the same configuration scores 5 of 17 against the baseline's 6. **Both optima are computed with the answer key and are bounds, not methods.** Sources: a0_baseline.json 7645e4746715a965, e005_results.json 07b714389697c6e5, achievability.json 770792d09e07978d.

Both optima are computed with the answer key. They are bounds, not methods, and they are not achievable by any retriever. One further qualifier: four of the five optimum episodes are *prior probe exchanges*, and this probe's earlier answers were largely wrong. An item counts as available if its text appears, however wrong the response surrounding it.

8.2 The candidate pool binds first, and structurally

The deployed path pre-filters the store to a 34-episode shortlist before any selector runs. **Neither art-domain contributor is in it** — they sit at cosine ranks 50 and 86 of 119.

That is not a measured comparison; it is a fact about the shortlist's contents. **No selection rule of any kind can reach the art domain from a shortlist that contains no art episode.** The constraint binds before any objective is evaluated, which is why it is first.

With the selector held fixed and only the pool widened, the same configuration moves from 5 of 17 across 2 domains to **12 of 17 across 4**. Dropping only the 19 lowest-cosine episodes of 119 costs three facts, the entire art domain, and all overlap with the known optimum — even though four of the five optimum episodes survive the cut. The selector clusters over its pool, so removing the tail reshuffles the objective rather than simply removing options.

This is the operational instruction with the widest reach: do not prune the candidate pool to control cost. The reason is structural rather than comparative, and it is the paragraph above: a shortlist with no art episode cannot yield art at any setting. This programme did not run a controlled comparison of pruning against not pruning.

8.3 The objective binds second, and only after

Run on the *deployed* 34-episode pool, the shipped set-level objective scores **5 of 17 against the baseline's 6**. It is worse than what it replaces.

That single count — one run, one configuration — is what makes the order forced rather than tidy. Improving the selection rule before widening the shortlist is not merely less effective; on this store it was negative. A practitioner who reads §8.1 and reaches for a better objective first will reproduce that result.

8.4 A similarity floor binds last

After the pool is wide and the objective is set-level, one episode remains unreachable. It carries four monetary items, sits at cosine rank 112 of 119 with a query cosine of **0.0560**, and would need **0.225** to be selected. Only 20 of 119 episodes clear that bar. Across 146 swept configurations, **zero** selected it.

The tempting explanation — that it collided with a higher-scoring episode in the same diversity cluster — was tested and **refuted**: its cluster is never entered at all, so the diversity term was payable in full at every step. No reweighting closes a gap of that size. The programme shipped the configuration with the miss characterized rather than tuned away.

8.5 What the internal inversion does and does not mean

On this corpus's one enumeration probe, the four highest-cosine episodes carry none of its target facts, and the last needed item does not appear until rank 87 of 119. That is a striking number, and it has been tested externally, where it **narrows sharply**.

On 470 answerable LongMemEval questions, only **69 (14.7%)** place every piece of evidence below the top four. Median evidence-session rank is **2**; the 95th percentile is 23. The internal inversion is real and it is **not the dominant external pattern**.

Two further constraints on how far it travels. The same internal ordering places every needed item inside rank 2 on all eight targeted probes — so this is *one probe behaving unlike eight*, not an established property of query types. And a diagnostic built to test whether the planted vocabulary caused the inversion could not be completed: the prior rarity artifact scores only 6 of 76 fact-bearing episodes across three variants with no registered primary. That is a measurement-unit failure rather than a null, and the categorical claim it was meant to support has been withdrawn.

9. Packing priority is a causal delivery gate

Standing: REGISTERED-OFFLINE. The order in which a fixed candidate set is packed into a fixed budget decides what arrives. This was measured twice, once externally and once internally, holding everything else constant.

9.1 EC-002 — reversing the order on 500 external stores

The 500 stores from the external calibration run, replayed with **only the packing order changed**. The deployed order fills recency first, then similarity, then coverage. The counterfactual fills similarity first. No reader inference, no embedding call.

Outcome	Deployed order	Similarity-first	Gains	Losses
Any evidence session	109 / 470 (23.2%)	**261 / 470 (55.5%)**	152	**0**
All evidence sessions	34 / 470	137 / 470	103	0
Any exact answer turn	79 / 470 (16.8%)	196 / 470 (41.7%)	119	2
All exact answer turns	20 / 470	106 / 470	86	0

Plus 32.3 percentage points on the primary, with 152 gains and no losses. Figure 8. One qualifier travels with this: the deployed arm is a *reproduction under recomputed embeddings*, not a byte-exact replay — the original run's embedding cache was not retained, and that run is permanently unreplayable at bit granularity. Restricted to the 401 questions whose evidence was already in the top four ranks, recall goes from 96 to 248. The evidence was ranked correctly and then not delivered.

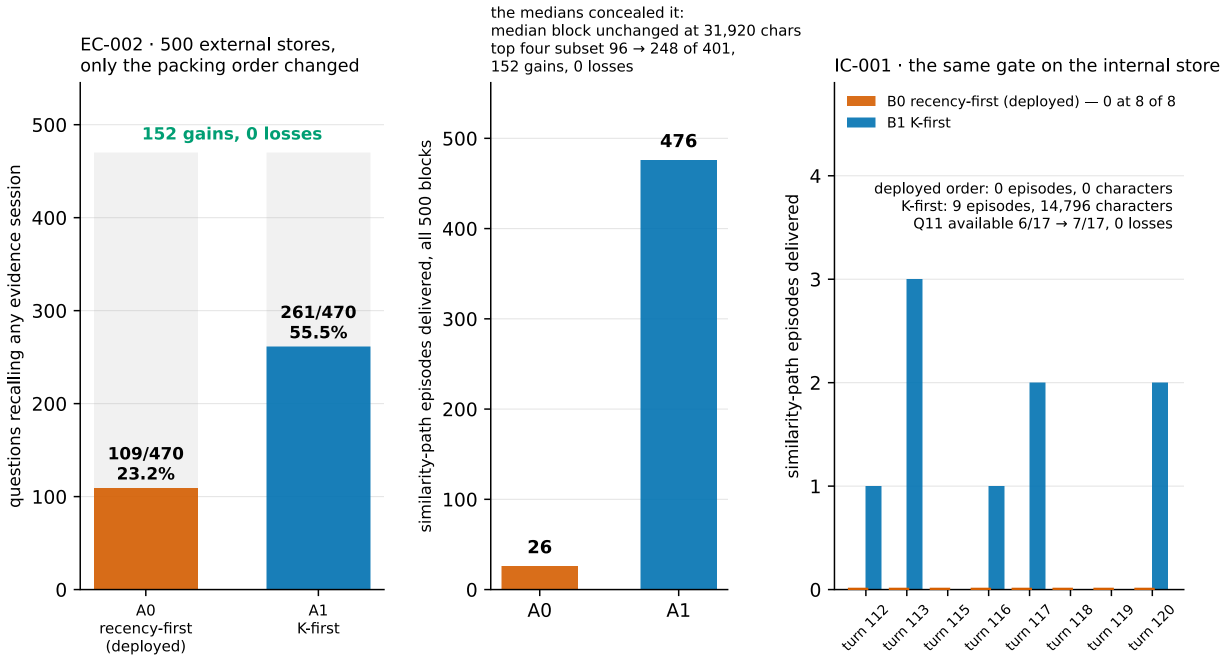

Figure 8. Packing priority is a delivery gate. *The evidence was ranked correctly and then not delivered.* Left: reversing only the fill order on 500 external stores moves any-evidence-session recall from 109 to 261 of 470, with 152 gains and zero losses, while the median delivered block stays at 31,920 characters and the median path composition stays 16 recency, 0 non-recency similarity, 1 coverage. The aggregate that moves is delivered similarity episodes, 26 to 476 — the medians concealed a twentyfold change in the tier the system exists to provide. Right: on the internal store the deployed order delivered zero similarity episodes and zero characters at 8 of 8 probes. §9.3 records that the live correction built on this was tested and rejected. Sources: paired_comparison.json 6f18b46f7316f8dc, paired_comparison.json 5dad3eabfac3df39, path_split.csv d26440d1476e8923.

The medians concealed all of it. Block size stays at 31,920 characters; the median composition stays 16 recency episodes, 0 non-recency similarity episodes, 1 coverage episode. What moves is the aggregate: delivered similarity episodes rise from **26 to 476**. A summary statistic that looked stable across the change was hiding a twentyfold difference in the tier the system exists to provide.

Residual, stated because it bounds the fix: 209 of 470 still recall no evidence session under the better order.

9.2 IC-001 — the same gate on the internal store

Both orders replayed over frozen candidate identities; zero inference, zero embedding, no vector re-derived.

Under the deployed order, the similarity path delivered zero episodes and zero characters at 8 of 8 probes. Every internal number this programme published for that configuration sits behind a starved fill order. Reversing it delivers 9 episodes and 14,796 characters. The enumeration probe moves 6 to 7 of 17; targeted probes move 14 to 18 of 21, with four gains and zero losses. At the enumeration probe the reversed order delivers 12 episodes in 31,863 characters against 8 in 31,946 — four more episodes in 83 fewer characters.

9.3 What was done about it, and why that is not a contradiction

The correction was tested live and **rejected**. A registered comparison scored the similarity-first arm at 7.0 against the deployed arm's 8.0, and the registration's own bar fired. The correction is not adopted.

Those two facts sit together honestly. The suppression is confirmed — it is an offline count that reproduces on replay — under recomputed embeddings for the external arm, per §9.1 — and §13.1's noise band does not touch it. The live comparison that would have justified adopting the fix went the other way, and its −1.0 margin is *inside* the band and therefore **not demonstrated in either direction**. The programme's registration forbids citing the band to revive the rejected correction, and this paper does not. What is established is that packing order is a delivery gate. What is not established is that reversing it improves answers.

10. Cost and the operating envelope

Standing: DESCRIPTIVE. These come from a 1,000-turn endurance run, not the 121-turn corpus §8 uses. Nothing here establishes a §8 result at 1,000 turns, and nothing in §8 establishes these at 121.

Three things could grow as a conversation lengthens. Only one binds.

Delivered context is bounded, because it is enforced. Replaying 1,000 committed episodes through the library at a 32,000-character budget, the delivered block breaches the budget on **0 of 1,000 turns**, and its 95th percentile moves **+18 characters** across the final five 100-turn buckets. The same block also **truncates on 895 of those turns**, dropping up to 70 episodes and wanting up to 65,864 characters. It is bounded because a ceiling binds during selection, not because demand is small. Both readings belong together; the first alone would be the kind of surrogate this programme keeps catching.

Disk is cheap. 4,743 bytes per turn at the margin, 86% of it embeddings. About 48 MB at 10,000 turns. Nothing there ends continuous operation.

Latency binds. 190 ms at 1,000 candidates, with clustering taking 81% of it and that share rising from 37%. The measured exponent is 1.25 over 50 to 1,000 candidates. The stated horizon is comfortable to a few thousand episodes and unusable in an interactive loop somewhere before 10,000. Figure 11.

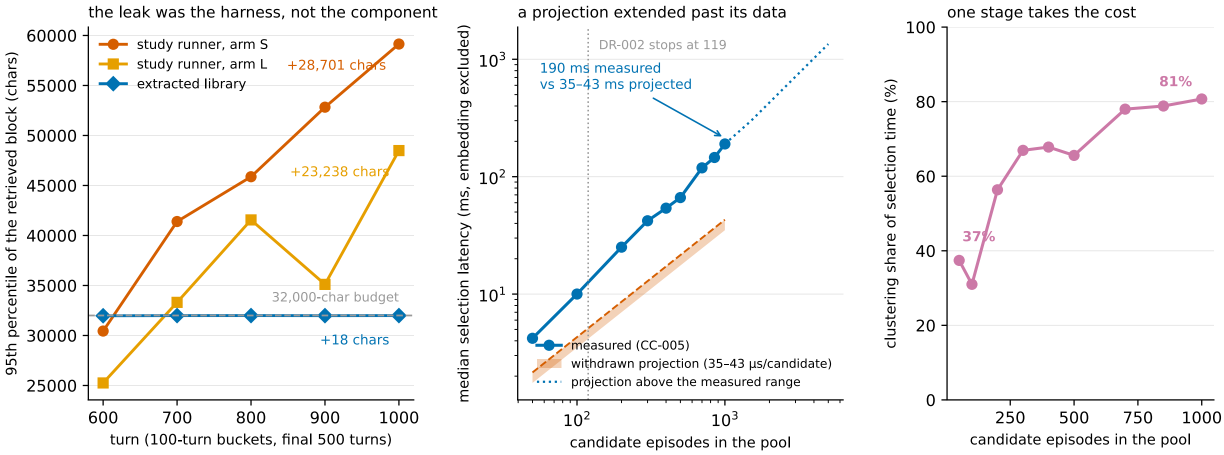

Figure 11. Growth and cost. *The growth belonged to the harness; the cost was five times its projection.* Left: the 95th percentile of the retrieved block per 100-turn bucket over the final 500 turns of the 1,000-turn run. In the study runner the block rises 23,238 characters in one arm and 28,701 in the other, still setting records in the last bucket; replayed through the extracted library at the same budget it moves +18 characters and breaches the budget on 0 of 1,000 turns — while truncating on 895 of them, so it is bounded because enforced, not because demand is small. Right: measured median selection latency against candidate count, with the withdrawn projection drawn as the range it actually published rather than a single point. 190 ms measured at 1,000 candidates, exponent 1.25 over 50–1,000, and clustering's share rising from 37% to 81%. Values above 1,000 candidates are projections, drawn dashed. Sources: dx002_results.json f8ab79ab041cb3e3, ge0_growth_gate.json 350fe20bbc3beed9, latency_curve.csv 0d2b8075ff5a971f, growth_measurement.json 20d65a018e28b03f, latency_components.csv 82bbbe38e9423d00.

That last number corrects a published projection of this programme's own, and the correction is instructive: the withdrawn estimate of about 40 ms at 1,000 candidates came from extending a curve **84 times beyond its last measured point**. Measuring instead gave roughly five times the projection.

These milliseconds are one machine's. The runtime is llama.cpp with a 27B generation model at UD-Q6_K_XL, one slot, fixed seed, speculative decoding disabled, and Qwen3-Embedding-0.6B over SQLite with sqlite-vec. Selection timings exclude embedding entirely, because query and episode vectors are already resident. Only the exponent and the clustering share plausibly transfer.

The obvious optimization is the one thing not to do. Keeping the pool small by dropping low-similarity episodes is what §8.2 shows removes a domain from the shortlist outright. So retention is unbounded by policy, the trimming knob carries an unsafe_ prefix with the finding in its docstring, and the horizon is stated rather than engineered around.

Growth belonged to the harness, not the design. In the original study runner the retrieved block rose 23,238 characters in one arm and 28,701 in the other, still setting records in the final bucket. Replayed through the extracted library at the same budget it moves +18. The leak was the runner's.

11. What was removed

Each mechanism below was built, measured, and closed by the result beside it. They do not correspond one-to-one with the studies: several fell to the same study, and some outlived the study that first weakened them. Figure 7 places each against the bar it had to clear.

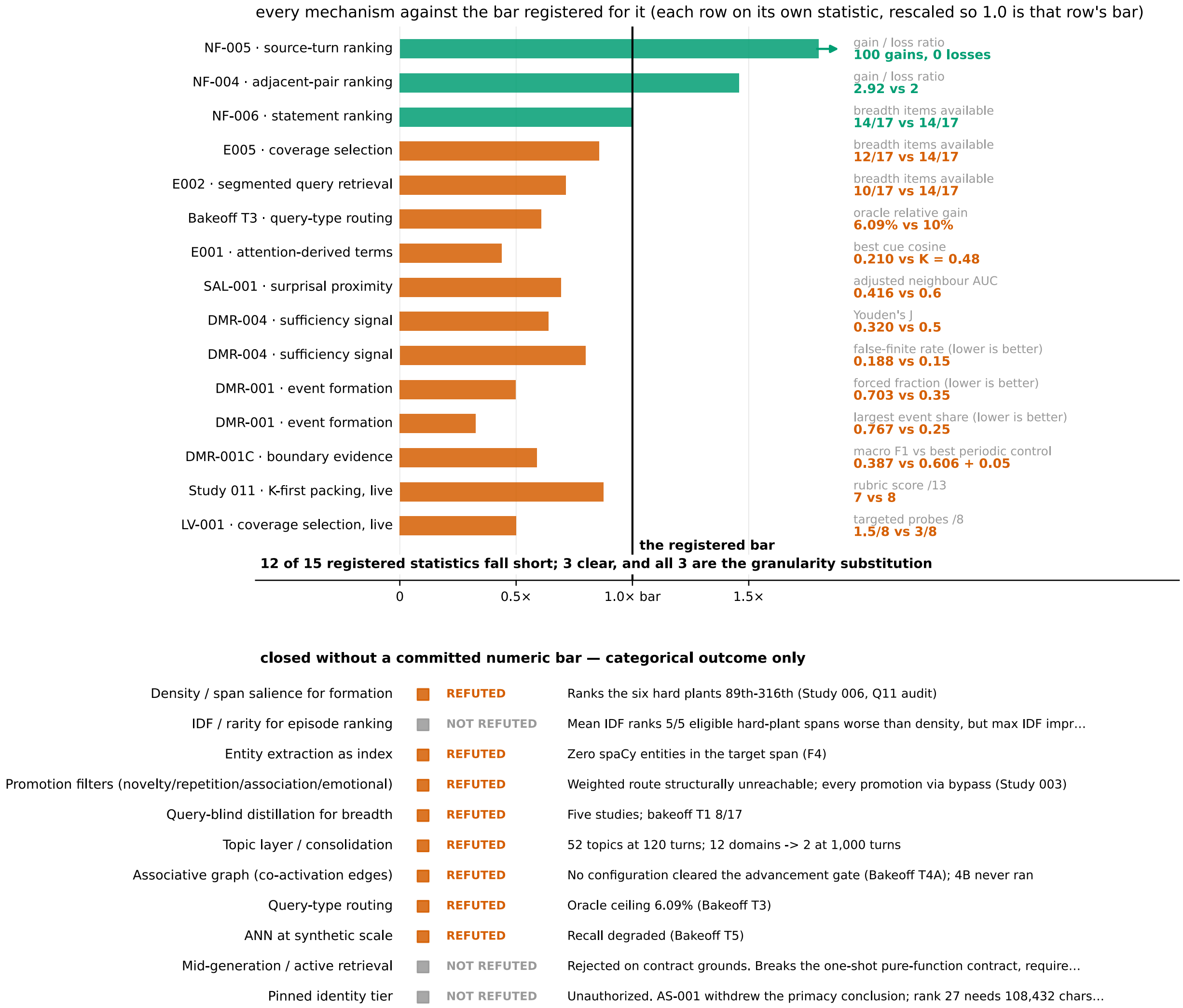

Figure 7. Every mechanism against its own bar. *Twelve mechanisms fell short of the bar each had to clear; the three that cleared are all the same family.* Each numeric row is rescaled so 1.0 is that row's own registered bar, which is the only way results measured in incompatible units belong on one axis. Fifteen rows have a committed numeric bar; eleven more have none and are drawn categorically with the reason parsed from the ledger rather than assigned a number. The three rows above 1.0 are NF-004, NF-005 and NF-006 — the granularity family. **This is the figure that answers why the survivor survived: not because it was designed better, but because it is the only thing that cleared its gate.** Sources: RETRIEVAL_MECHANISM_LEDGER.md 6c3875cc4f63046d, retrieval_bakeoff_report.md 681208a78d12f591, tier3_results.json 8c229f7b70de9a70, e001_results.json a996ecd881ae52aa, gate_report.json 1a43d8a5345188af, gates_holdout.json 13dd30919acd91bf.

Removed	Killed by
Distillation and “dreaming”	Five studies; query-blind selection cannot anticipate a later query
Promotion filters	The weighted route was arithmetically unreachable; every promotion came via bypass
Topic layer and consolidation	52 topics for one 120-turn conversation; 12 domains collapsed to 2 at 1,000 turns
Associative graph from co-activation	No configuration cleared its advancement gate at any of eight registered edge sets across three depths
Query-type routing	Oracle ceiling 6.09% against a registered 10% build threshold
Approximate nearest-neighbour search	Recall degraded at synthetic scale

Removed	Killed by
Query segmentation	Improved its matched-budget baseline from 6/17 to 10/17 and still failed its locked 14/17 bar
Attention-derived term selection	Run as an oracle over 714 candidate cue rows: 0 reached the retrieval threshold
Entity extraction as primary index	Zero entities in the target span
Density for formation	Ranks the six hardest planted facts between 89th and 316th
Rule detection and persistence	118 false rules by turn 200; failed at 1,000-turn scale
Surprisal-based capture	SAL-001, sealed holdout: AUC 0.416 against a 0.60 bar, five of six strata below chance
Deterministic adaptive stopping	DMR-004, sealed holdout: Youden's J 0.320 against a 0.50 bar
Absolute-threshold event segmentation	DMR-001, sealed holdout: forced fraction 0.703 against a 0.35 bar

Two results from the numbered arc carry the argument.

Write-time selection cannot anticipate a query. Studies 003 through 007 are five attempts to decide, at write time, what deserves remembering. Each optimized a proxy satisfiable without the property it certified: record count for information, novelty for importance, density for factual value. The terminal diagnosis is specific — density, the best write-time salience signal available, ranks the six hardest planted facts between 89th and 316th. These are rare technical phrases whose component words are common.

Moving selection to query time did not recover it. A registered bakeoff made that repair its central premise and refuted it. The best registered 32,000-character retrieval block surfaced **8 of 17** target facts, below the 11 of 17 the formation era reached. All 17 were present in the raw store. Retrieval did not find them.

What remains is §3.1's four components. It contains none of the mechanisms above.

One item is retired rather than solved. The component emits no absence signal on any of 500 external questions, while the fixed reader correctly abstained on 17 of 20 registered abstention items. That does not give the component an absence detector, and it does not establish reader reliability across models and prompts. It shows that component-level detection is not necessary for end-to-end abstention *under this tested reader*, which is why the requirement is retired at the component level rather than marked solved.

12. Self-audit and corrections

Everything internal in this paper rests on this programme's own measurements, scored by its own raters, against its own rubric. The reason to extend it credit is that it audited itself and published what it found. ERRATA.md holds 19 corrections; all were caught by gates the programme wrote. This section gives the ones that change how a reader should read the rest.

12.1 The scoring audit removed the programme's only success

A blind re-scoring of 222 committed items across nine studies changed 19. One study's headline arm fell from 13.0 to 8.5, because a truncated reasoning block had been credited as a complete response. Study 001 lost the programme's only VALIDATED verdict and became PARTIAL.

The residual estimate is extrapolated, and its precision is reported with it: 3 disagreements in a 26-item control sample projected across 143 unreviewed items gives 16.5 expected errors, but the 95% Clopper-Pearson interval on 3 of 26 runs from **about 3 to about 43**. An earlier version of this paper reported "about 20" without the interval. That is the defect, not the estimate.

12.2 Every study on record ran over its stated budget

Character budgets were charged against source text rather than the complete serialized block — tags, metadata and separators excluded. Correcting it moved published numbers by up to 68%: two blocks reported at 31,991 and 31,847 characters against a 32,000 budget were actually **53,726 and 53,839**. The budget was violated, not saturated, and the scaling conclusion derived from the undercharged values is withdrawn. Every figure in this paper uses the corrected accounting.

12.3 The same query text returns different vectors

Reproducing a retrieval result requires reproducing the *embedding call shape*, not only the query text. The same embedder, given the same text solo rather than in a batch of nine, returns a vector agreeing to cosine

0.999837 — with a largest component difference of 0.217, and that difference **flips 6 of 146 committed payloads**.

This is why §13.4 does not treat embedder sensitivity as unmeasured. A perturbation far smaller than a model change already moves 4% of results. The library now asserts a sentinel vector hash on every store open, so a mismatched embedder fails loudly instead of silently returning different answers.

12.4 The tier called a recency window was not one

Three different rules have carried that name in this programme, and only the one in the extracted library is a window.

The rule every live study through the endurance run actually ran sorts the whole store by delivery history, and `retrieve()` refreshes what it just delivered — so the block re-selects itself. Tie-breaking on ascending turn number means it settles on the oldest episodes and cannot leave. **From turn 11 onward it delivered source turns 1 through 9 plus the immediately previous turn, for 111 consecutive turns.** Mean overlap with a true window of the same size is **0.205**; 111 of 120 episodes were delivered exactly once.

Two consequences. The programme's cleanest architectural contrast did not compare a memory tier against a recency baseline — it compared it against nine frozen episodes and one recent one. And **nothing in this programme establishes what a correctly implemented window would score, in either direction.** No arm ever ran one.

12.5 A diagnostic written to catch surrogate failures nearly committed one

The clearest methodological lesson here. A granularity study reported 49 gains and zero losses on evidence delivery. Its own preflight surrogate audit — the check that asks *can this pass while the property it certifies is false?* — then found that 94 treatment hits contained no answer-bearing episode at all. The measure was **session-touch**: it credited delivering *any* part of a session containing the answer.

Under the strict measure the result does not shrink. **It reverses**: 388 to 351, with 26 gains and 63 losses. The claim was withdrawn.

Two smaller failures sit inside the same study, and both are unit errors in a study whose entire subject is units. A first pass compared evidence *episodes* against evidence *sessions* and reported a 45-item regression that did not exist. A denominator counted five never-ranked items as misses.

Session-touch is not used anywhere in this paper. On LoCoMo development it reports 40 gains and zero losses where the strict measure reports 44 gains and **9 losses** — it hides every strict loss.

12.6 What the gates missed

A gate is only trustworthy if its tested population could have existed. Three cases here where one could not:

- A targeted no-regression gate keyed on `(turn, item)` while its requirement keyed on `(question, turn, item)`. The condition was **unsatisfiable by construction, for any selector**. Correcting it turned a recorded failure into a pass.
- A cluster-floor study stopped as inert — and the first execution reached that branch only because its evaluator searched for domain labels that did not exist. The invalid artifact is preserved and corrected by a standalone amendment.
- Fourteen repository integrity gates failed unconditionally for months because constants were recorded under one line-ending convention and checked under another. The cost was fourteen gates that could not have detected real drift, and one real drift was sitting behind them.
- A fifteenth was found while this paper was being written, on the sealed holdout itself. The constant binding NF-004's LoCoMo corpus-lock record had never matched the file it names: that file has one revision and one hash, and the expected value matched nothing in the repository. Its two sibling constants verify clean, and the corpus lock's content is independently checkable and intact, so **no NF-004 number moves**. What moved is the gate's status — for its whole life it could not have detected a real change to the record it guards. It was recorded here while still unfixed, then corrected separately once the decision belonged to someone entitled to make it; `ERRATA.md` carries both halves in order. The correction makes the gate bind from now on. It does not retroactively validate the period it was inert, and NF-004's result never rested on it: that rests on a replay hash equalling its committed value and a vector seal reading 2,749 of 2,749 with zero misses.

13. Limitations

Each item names what would settle it. This is the complete list rather than a restatement of scope caps already given at the claims they bound.

13.1 The instrument's band is 3.0, and most scored comparisons here are below it

Every *scored* comparison in this paper is a single run at a fixed seed. That was recorded as a missing variance estimate until the estimate was made.

Five replicates of the deployed configuration — identical corpus, settings, seed and standing runtime, run back to back in one server process — scored **11.0, 8.0, 8.0, 8.0 and 8.0**, in that order. Max minus min is **3.0**, against a decision rule committed before the replicates ran.

It is not a spread but a switch, and the run order is the finding. Replicates two through five are byte-identical across all 121 turns. **The one that diverges is replicate one — the first run in a fresh server process**, the only one to meet an empty slot. It diverges at turn 1 from a byte-identical 757-byte prompt and never re-converges.

Stated that way it is sharper than "one of five was odd", and it reaches further: every scored run in this arc that began on a cold server sits on the divergent side of that switch, and no study in the arc pinned process state (§13.2). Rater disagreement was measured separately and is near zero, 64 of 65 items unanimous, so this is run-to-run variation and not scoring noise.

Figure 9 draws the band across every scored verdict in the arc. Applied uniformly and in both directions, **three of this arc's four scored verdicts fall inside the band and are not demonstrated**: the memory-tier contrast at 3.0, the live-validation targeted regression at −2.0, and the tier-isolation result at −1.0. *Not demonstrated is not refuted.* These may be real, and a single run per arm cannot say. One asymmetry is worth stating: the memory-tier contrast has **less** protection than the others, not more — its two arms ran hours apart and neither manifest records a server process id, so the process state of the arc's cleanest architectural comparison is unknowable from the committed artifacts.

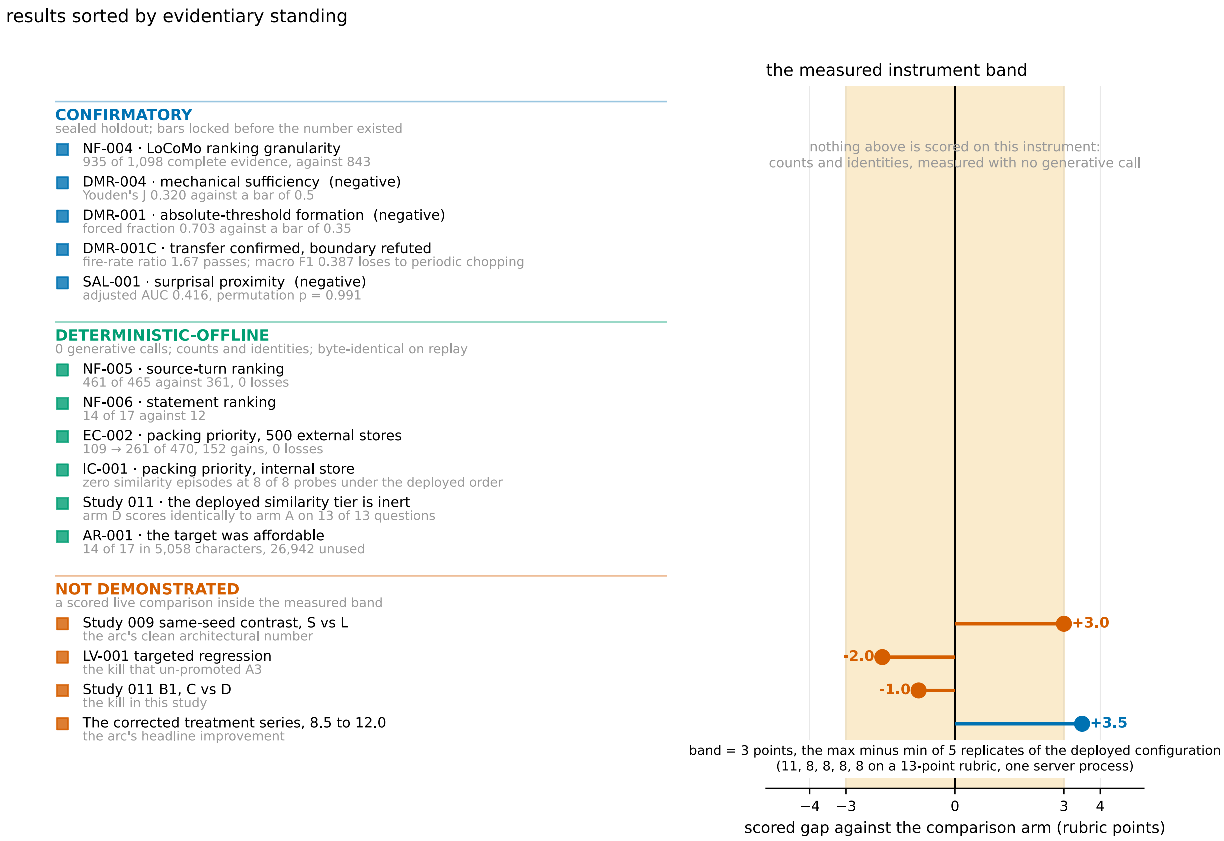

Figure 9. The standing ladder and the instrument band. *What the programme measured, sorted by what would make it believable.* Fifteen results in three tiers — confirmatory under a sealed holdout, deterministic and offline, and scored-live. The shaded region is the measured run-to-run band of 3.0 points, from five replicates of one configuration under identical corpus, settings, seed and runtime that scored 8.0, 8.0, 8.0, 8.0 and 11.0. It swallows three of the arc's scored verdicts: the memory-tier contrast at +3.0, the live-validation regression at −2.0, and the tier-isolation result at −1.0. The corrected treatment series at +3.5 sits outside it, and exceeding a band is not the same as being demonstrated. **Nothing in the top two tiers is touched by the band** — those are counts and identities rather than scores. Sources: band_verdict.json cb241ac01cbcde5f, verdict.json 83bd4bbbc4209a48, LV_001_report.md cd3dc4cdaa450002, SAL_001_FINAL_DESIGN.json 783cc8ad7f5796ac.

What the band does not touch is everything offline and deterministic: gate outcomes, delivery counts, character accounting, packing measurements, §6's sealed holdout, §7's granularity results, §9's 152 gains and zero losses. Those are counts and identities, not scores. *Settled by:* repeated runs at multiple seeds with process state pinned, which no study in this arc pinned.

13.2 The runtime is not bit-reproducible

The programme's standing rule requires a byte-identical seeded-prefix rerun. On this runtime that rule is satisfiable **between runs that share server process state** and **not** between a cold-start run and a warm-start one — and no study in the arc pinned process state. An identical 757-byte prompt at seed 5005, with one slot and speculative decoding disabled, produced responses diverging at character 79.

This bounds every scored claim and none of the offline ones. *Settled by:* pinning process state as a run gate.

13.3 Availability is not correctness

Every selection count here measures whether a fact was *present in the delivered window*, not whether the model answered correctly. This was the largest structural weakness of this paper's predecessor. **It has now been measured, and the weakness is real.**

A pre-registered live run compared the shipping configuration against the deployed baseline at one seed, fixing both reporting outcomes before any number existed:

Registered bar	Result
Does the offline advantage convert?	**WEAK.** A six-item offline gap became +1 correctly attributed item
No targeted regression, tolerance 0.5	**FAIL.** 1.5 of 8 against the baseline's 3.5 — a 2.0 shortfall, four times the tolerance

The second was registered as a kill. The configuration's status is **not promoted**.

The mechanism is legible. Asked for two formatting rules planted in the first two turns, the shipping configuration reported that it could not see the start of the conversation: the coverage objective had spent its budget on domain spread and stopped carrying the opening, which per-item cosine ranking had retained. Offline, that same configuration preserved 16 of 16 targeted items. **Preserving an item's availability and preserving the answer that depends on it are not the same property, and this programme had measured only the first.**

The finding that survives the band is not the score. Both arms fabricated confidently on the domain neither retrieved, one attributing a painting to the wrong artist while still producing both correct pigment terms — which a presence-only scorer credits as a hit. That is an identity, not a rating, so §13.1 does not touch it, and it is the more durable result of the two: a measure that counts an item as available when its text appears will score a confident fabrication as a success. The −2.0 magnitude is as unreplicated as the +1 and neither is demonstrated; the kill bar fired on a threshold registered before either number existed, and that stands independently. *Settled by:* the frozen-context reader study described in §14.

13.4 One runtime, one embedder, and positive evidence of fragility

One model, one quantization, one machine, one embedder. Whether §8's results survive a *different* embedder is unmeasured — but this is not a blank absence of evidence, and it would be convenient to call it one. §12.3 shows the *same* embedder under a different call shape flipping 6 of 146 committed payloads. A perturbation far smaller than a model change already moves 4% of results, so the reasonable prior is that §8's specific numbers are embedder-dependent. *Settled by:* rerunning the sweep under a second embedder.

13.5 External scoring is substituted, not official

The external calibration run's end-to-end scores — 20.0% on an equal-quota subset and 12.22% post-stratified — are **Codex-substituted integrity scores**, because the benchmark's pinned evaluator was unavailable and was replaced by a panel of hosted models with AI adjudication. A binding amendment forbids placing either number against published LongMemEval results, and this paper does not.

Two further boundaries. The corpus is a cleaned variant rather than the original histories. And the benchmark authors' LLM-assisted indexing and time-aware query expansion were available and **deliberately not adopted**, because they add generative calls to the memory path. Some of the gap between this component and published systems is that choice, and this paper does not claim the choice was free. *Settled by:* the official evaluator on the registered answers.

13.6 The internal corpus is constructed, and one probe carries the breadth claims

Every internal breadth number — 6, 12, 14 and 15 of 17, and the rank-87 reading — comes from **one enumeration question** at one turn. The corpus is constructed, and its planted vocabulary is a specific reason to suspect the ranking inversion is a property of the plant rather than of retrieval. §8.5 gives the external test, which narrows the claim sharply, and §12.5 gives the rarity diagnostic that failed to identify the mechanism. *Settled by:* multiple literal enumeration probes across external domains.

13.7 The endurance corpus is 84% repeats

The 1,000-turn run holds only **156 distinct user-assistant pairs**; 844 of 1,000 episodes are exact content duplicates. Any saturation reading from that run has a mechanical reason to saturate independent of the mechanism, and §10's growth and latency findings are the parts unaffected. That run's scores were outside the scoring audit and one of its arms was budget-noncompliant; they are not used in this paper.

13.8 AI raters, AI adjudicators

Scoring used three blind passes with registered adjudication triggers, but the adjudicators were subagents rather than humans, and the three raters were three models from one family — a departure disclosed in advance. Shared-family bias inflates apparent agreement, and inflated agreement *understates* a band. The control sample disagreed at 11.54%. *Settled by:* human adjudication of the same items.

13.9 Amendments exist after results

Twelve in the bakeoff alone. The programme records per amendment whether it preceded the result it affects, and applies a legitimacy test permitting corrections to measurement units and protocol contradictions while forbidding making a criterion easier once results are known. The record is published so a reader can disagree with individual calls.

13.10 LongMemEval is exhausted

Every item in that corpus has now been used by this programme. **No confirmatory claim is available from it again**, and any registration written today inherits that ceiling. The LoCoMo holdout in §6.1 is now the only sealed external evidence this programme holds for the granularity result, and **LoCoMo is now spent three times over**: NF-004 read the six holdout conversations, HH-001 read them again, and HH-002 read all ten. Both studies in §5 are REGISTERED-LIVE for that reason and cannot become confirmatory. No sealed external corpus remains to this programme.

14. Conclusion

Eleven pre-registered efforts produced one architecture worth keeping, one externally confirmed rule about how to use it, and a measurable account of why the rest did not work.

For a practitioner, three things transfer.

Score a candidate by its own text, not by its parent's. This is the substitution with sealed external confirmation — 843 to 935 of 1,098 on withheld conversations, reproduced at 361 to 461 of 465 on a second corpus and 12 to 14 of 17 internally. The mechanism is candidate size: a 2,550-character episode averages away the 298-character turn that answers the question.

And do not read that as "finer is always better", because this paper's own data refutes it. On the same LongMemEval corpus, moving the ranking unit from the session to the episode **loses 37 items**, 26 gains against 63 losses (§7.3). Turn ranking beats episode ranking; session ranking also beats episode ranking. The episode loses from both sides, which no monotone rule about fineness can produce. The confirmed result is the specific substitution above, on units of roughly 240 to 300 characters. Where the useful unit sits for a different corpus is not something this programme measured, and §7.5 refutes the scope condition proposed to predict it.

Do not prune the candidate pool to control cost. On this store the deployed shortlist contains no representative of one of four domains, so no selection rule reaches it from there. That is a fact about the shortlist's contents rather than a measured comparison, and it is the sharper reason to leave retention alone.

Check whether your memory layer needs generative calls at all. Every component this programme removed was one that required them, and what is left is replayable and provenance-preserving as a consequence. Mem0, Zep, Letta and Graphiti each spend at least one generative call on this layer by their own published descriptions. One of them was run here. On 300 questions, one reader and a matched budget, the layer that spends nothing answered more of them than the layer that spent 1,646 calls and 284 minutes — and reached its store 41 times faster, in a sixth of the space, having dropped none of what it was given.

Three of the five sealed results were negative, and they are why the list above is short. A deterministic stopping controller reached Youden's J of 0.320 against a registered 0.50. An absolute-threshold event segmenter closed 52 of 74 events on its size cap against a 0.35 bar, and its relative successor then lost to chopping every four episodes regardless of content. A surprisal-proximity capture signal scored AUC 0.416 against a 0.60 bar, below chance in five of six strata. Each was pre-registered against a sealed holdout, and each closed a line this programme wanted to keep.

For a researcher, the useful part is the decomposition and the order inside it. Retrieval failure here was not one thing. The candidate pool decided what could be seen, the objective decided what was worth taking, and a similarity floor decided what was unreachable at any weighting. The order is forced and structurally so: one domain has no representative anywhere in the deployed shortlist, so no objective can recover it from there. Widening the pool is not the first fix because it measured better — it is first because the alternative is impossible. Separating the three required a per-fact known optimum on the same store, a measurement costing an answer key and exact cost accounting, and buying a sharper question than an end-to-end score.

What this programme has not shown is that its *mechanism* results carry through to a reader. §5 measures answers directly and this component wins there; what remains unestablished is the link between the two — that the availability gains §7 confirms are what produce those answers. Availability and correctness were measured moving in opposite directions once, and that result stands unrescued. The next decision-relevant experiment is therefore not another retrieval study. It is a reader study over two already-frozen contexts — whether the 14-of-17 context causes a reader to use more correct facts than the 12-of-17 context, under an identical prompt and runtime, scored as a 17-bit fact-use vector rather than on the

13-point rubric whose 3.0-point band cannot resolve a two-item contrast. That design is prepared and unregistered. The internal corpus is exhausted, so its maximum evidential status is characterization, and the value is causal interpretation rather than fresh-corpus confirmation.

The programme's summary of ten studies and one bakeoff is that, at the one probe where it was measured item by item, the model used all ten available facts and invented none. That is a statement about one probe and it does not generalize: §13.3 records the counterexample, where both live arms fabricated confidently on the domain neither of them retrieved — one of them producing the correct pigment terms attached to the wrong painter, which a presence-only scorer credits.

So the failures measured here were delivery failures, and delivery turned out to be a selection problem sitting on top of a candidate set already narrowed by the wrong rule and scored at the wrong unit. What happens when delivery succeeds and the reader still gets it wrong is the part this programme has not measured.

Appendices

A. Evidence spine — paper/notes/EVIDENCE_SPINE.md. Every number in this paper with its artifact, its SHA prefix where one exists, and its standing under §4.1's taxonomy. Built before the prose, so the draft was assembled from it rather than checked against it afterwards.

B. Withdrawn claims — paper/notes/DO_NOT_WRITE.md. Thirty-five sentences this programme has published and then corrected, each with its replacement, drawn from ERRATA.md and two adversarial review cycles. A rewrite is when withdrawn claims come back, because they are usually the cleaner sentence.

C. Claim gates — scripts/check_paper_002_claims.py. Two automated checks: every numeric literal in this paper must appear in the evidence spine, and no value on the machine-checkable superseded list may appear at all. Both pass at the committed revision. Neither proves a claim is right; they catch the two failure modes this repository has actually committed.

D. Corrections index — ERRATA.md, 20 entries, cross-referenced from §12. It includes one entry that was wrong when first written and is superseded in place by its own reversal, with the original text kept so the mistake stays visible.

E. Study reports — experiments/. Each carries its pre-registration commit SHA, its preflight record, its gate outcomes, and its own boundary section. Where a report and a summary disagree, the report is authoritative; this paper was written from the reports.

F. Amendment record — per-study amendments/ directories, each a standalone file recording whether it preceded the result it affects.

G. Reproduction — paper/REPRODUCTION.md and paper/reproduce_headline.py. The script rebuilds the 5,058-character exact optimum from the committed turn log through the installed library and checks its length and SHA-256 against the committed values, verified in a clean virtual environment containing only episodic.

Stated plainly, because it is a limitation and not a feature: **the selection results cannot be reproduced this way.** They need per-episode embedding vectors, which live in a store that is gitignored and was never committed, and regenerating them needs the pinned embedder, which is not in the repository either. One number in this paper is reproducible from committed data alone, and it is that one.

H. Review record — paper/reviews/. Two adversarial cycles against this paper's predecessor (sixteen objections and ten, all accepted) and a slop audit. The constraints they established carry forward into this document and are listed in appendix B §4.

I. Pipeline log — paper/notes/PIPELINE_LOG.md. The academic-research-skills stages this rewrite ran, and the verdict returned at each checkpoint.

J. Competitive landscape — paper/notes/COMPETITIVE_LANDSCAPE.md. Published results for the systems named in §2, with citation verification status and the boundary. §4.1 of that file records what HH-001 changed: Mem0 was run, so a comparison to **Mem0 as run here** is measured; every other system in the file is still published-only, and a comparison to Mem0's published score is still forbidden.

Provenance of this document

The paper is generated, not authored. This Markdown file is the only place a claim may be edited. paper/figures/ is a build output of scripts/generate_paper_002_figures.py, which reads every plotted value from a committed artifact and records the SHA-256 of each input in paper/figures/figure_manifest_002.json. The PDF is a build output of scripts/build_paper_pdf.py. Hand-editing a figure, the manifest, or the PDF is a defect rather than a shortcut.

If this file and a figure disagree, this file is right and the build script is broken. If this file and a study's pre-registration disagree, **the pre-registration governs** — that conflict is a defect to be flagged, not reconciled silently.

Supersedes PAPER-001. The predecessor's structure led with the negative results and reached its sealed-holdout confirmation in passing. Two numbers changed in this rewrite; the ordering and the standing labels did. Six stale cross-references in the predecessor were found during the rewrite audit and are listed in appendix B §6.